

# The best defensive strategies: two centuries of evidence<sup>1</sup>

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## Abstract

We examine downside protection – or defensive – strategies over more than 220 years of global financial history – covering many years where traditional equity-bond portfolios suffer and across a wide range of economic scenarios and historical regimes. Traditional defensive equity factors – low-risk, quality, and value – consistently provide effective downside protection, whereas gold and put options prove less drawdown- or cost-effective. Our long-run evidence shows that multi-asset defensive strategies, particularly a return-enhanced version of the Defensive Absolute Return (DAR) portfolio introduced by Cavaglia et al. (2022) and trend-following, provide the most effective downside protection. DAR and trend-following are complementary across tests by diversifying each other across stages of drawdowns. Investors can improve the defensive properties and improve total portfolio outcomes of traditional portfolios by considering the deep sample evidence on defensive strategies provided in this paper.

**Keywords:** Defensive strategies, Downside protection, Defensive Absolute Return, Trend following, Bad-times, Recession, Gold.

**JEL Classification:** E32, E44, G01, G11, G12, G14.

**EFM Classification:** 370, 450.

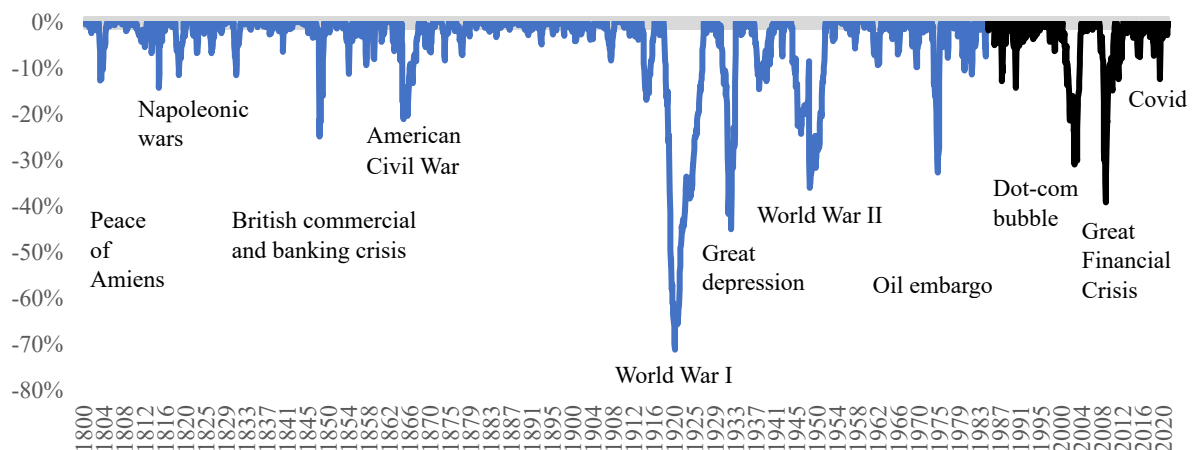
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## 1. Introduction

Traditionally, investors rely on equity and bond market premia to drive returns on their portfolios. A common reference is the 60% equity – 40% bond portfolio, which in the long run harvests the equity and bond market risk premiums. This portfolio has delivered compelling returns of 7.0% per annum over about the last 222 years (see Baltussen et al., 2023). However, over shorter periods the 60/40 portfolio can exhibit substantial volatility and drawdowns. Figure 1 shows the historical drawdowns on the global 60/40 portfolio. As shown, drawdowns have been large, with a maximum loss of 71.2% in the period 1800 to 2021, thus posing material risk to investors.

**Figure 1: Historical drawdowns of the global 60% equities & 40% bonds portfolio**



Note: Shown are historical drawdowns of the global portfolio investing 60% in equities and 40% in bonds from 1800 to 2021. For each month we show how far the cumulative performance of this portfolio is below the historical maximum of the cumulative performance. A drawdown period is completed when a new high watermark is reached. The black line indicates the Harvey et al. (2019) sample while the blue line indicates the out of sample period. There are many drawdown periods, with the largest after World War I with 60/40 losing –71.2% from October 1918 to November 1926 and the second largest with 60/40 losing –45.0% during the Great Depression from October 1929 to June 1933.

For the short-term investor the risks are apparent. For the long-term investor, this risk materializes in a reduction in longer term wealth accumulation. Volatility drag, or variance drain, pertains to the concept that when an investor loses 10%, (s)he needs to earn 11% to get back to zero ( $1.1 \times$  the loss percentage). When an investor loses 50%, (s)he needs to earn 100% to get back to zero ( $2.0 \times$  the loss percentage). Consequently, investors have always been looking for a reduction in volatility drag, or downside protection, and practitioners and academics have proposed various solutions. For example, the topically named work ‘Drawdowns’ by van Hemert et. al (2020) evaluates the practice and challenges of drawdown management. The comprehensive study by Harvey et al. (2019) examines a range of downside protection strategies, including equity index put options, gold, trend-following strategies, and the quality factor in equities. They conclude that short-dated options are highly effective but also prohibitively costly, while holding safe-haven U.S. treasury bonds is unreliable due to the variability in the equity-bond correlation. Instead, they conclude that the best downside protection is provided by trend-following strategies and the quality factor in equities.

As a limiting factor, studies evaluating downside protection rely on relatively short samples, and therefore base conclusions on a limited number of drawdowns. The Harvey et al. (2019) sample spans the 1985 to 2018 period. Over this 34 years sample period of data *eight* equity drawdowns occurred where the S&P 500 lost more than 15% from its peak – excluding

many other major equity market drawdowns before and the more recent COVID-19 market crash of 2020. Figure 1 shows that over this period only *two* drawdowns of more than 15% in the global 60/40 portfolio occurred. Ideally, downside protection strategies are examined over a deeper sample so that as many as possible drawdown observations are included.

In this paper we address this challenge by examining downside protection strategies utilizing the deepest sample we could construct reliably, covering over 220 years of returns. For constructing our sample, we combine the data utilized in multiple ‘deep sample’ studies that construct historical datasets of asset and factor returns. These include Baltussen, Swinkels, and Van Vliet (2021), Baltussen, Swinkels, Van Vliet and Van Vliet (2023), Baltussen, Van Vliet, and Van Vliet (2023), and Wahal (2019). Our final dataset starts in 1800, is at the monthly frequency, and provides us with as much equity or bond market drawdowns as reliably possible. Over this sample, we extend the number of global 60/40 drawdowns substantially – for example drawdowns of more than 20% increase from two to *eight* – as highlighted in Figure 1.

We consider a range of downside protection – or defensive – strategies documented in the literature. These include trend-following, gold, safe-haven bonds, low-risk, value, and quality equity factors (albeit over a shorter sample period), treasuries, put options, and the more novel downside protection strategy recently proposed by Cavaglia, Hua Fan, and Wang (2022). With a plethora of factor strategies out there, they propose a downside protection portfolio that goes long and short in factor strategies based on their historical correlation estimates with the 60/40 equity bond portfolio. They find that these historical correlation estimates are good predictions for future correlations. They start with a broad set of long/short strategies, investing in single name equities, futures on equity, bond and commodity markets, and currency forwards. At the end of each month all factors are sorted based on their updated historical correlation with the 60/40 portfolio. The lower the correlation, the higher the rank. The downside protection portfolio is constructed by going long the top tercile ranked factors and going short the bottom tercile factors. The resulting portfolio has a strong negative correlation with the 60/40 portfolio. We will call it the Defensive Absolute Return portfolio (labelled ‘DAR original’).

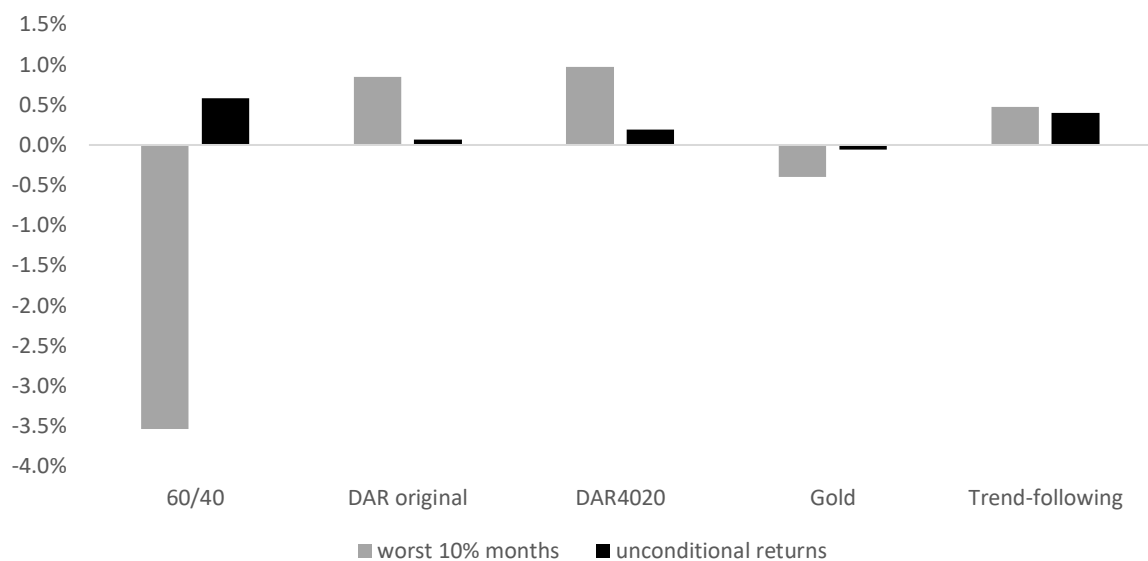
Our findings are as follows. First, over our sample of up to 222 years equities, bonds, and factors deliver attractive returns. Second, factors return correlations to 60/40 are persistent (i.e., past correlation estimates are predictive of future correlations). Third, examining key 60/40 down-market states, deeper tail drawdowns, and bad-times scenarios reveals traditional equity factors (i.e., low-risk, quality, and value) provide effective downside protection, but DAR original and trend-following provide even more effective drawdown reduction. For example, in the 10% worst 60/40 months these strategies return respectively 0.8% and 0.5% per month, and they are effective 68% and 58% of the time. This finding aligns with Hurst, Ooi, and Pedersen (2017), who show that trend-following strategies have performed especially well across various recessions and crisis periods since 1880. In contrast, gold provides on average negative excess returns of –0.4% in the worst 10% 60/40 months, delivering in the IT bubble and the Global Financial Crisis, but starting in 1800 it “fails” our out-of-sample test. Similarly, put options are sub-optimal as puts are expensive and come with a negative long-run performance, as also shown in Harvey et al. (2019) and Ilmanen et al. (2021). Third, we propose a return-improving DAR approach that provides better full-cycle characteristics (labelled ‘DAR4020’).

Harvey et al. (2019) build the case for only considering protective strategies that have long-run positive gross returns. These returns can be used to pay for trading costs and support the overall protective profile. As DAR original goes long and short factors in equal measure, its long-run returns are close to zero. Considering that correlations of factors are more predictable than returns, we propose an alternative portfolio that is long 40% factors with the most negative correlation, and short 20% factors with the most positive correlation

(DAR4020). We show that this alternative approach combines effective downside protection with positive long-run returns. It has improved protective characteristics compared to DAR original, and due to the factor premia, it has a long-run positive return. In the 10% worst 60/40 months DAR4020 returns 1.0% per month and it is effective in 67% of these worst months.

Figure 2 summarizes our primary results. Shown are the unconditional average returns for several downside protection strategies and the conditional average returns over the worst 10% of 60/40 months since 1800. Gold returns negatively in down states, while the DAR and trend-following approaches return positively. On average DAR4020 provides the highest protection in the worst 10% of 60/40 months, while trend-following strategies provide less average protection, but have the advantage that they make higher return in ‘good’ states, translating into larger average unconditional return. In the paper we show these findings are robust across shorter subsamples, including more recent ones, although we believe the more limited number of down states limit the impact of shorter sample inference.

**Figure 2: Downside protection strategies performance in worst and all 60/40 months**



Note: Shown are per defensive strategy the average performance in all months (“Unconditional returns”) and the average performance in the 10% worst months for the 60/40 equity/bond portfolio. All defensive strategies are full sample scaled to 5% to make them comparable. The sample period is 1800-2021. 60/40 is funded with the other strategies in excess of cash.

Next, we directly compare the effectiveness of the two most effective downside protection strategies: DAR4020 and trend-following. We find that simple trend-following strategies, on average, provide less protection at the start of a drawdown but pick up speed the longer the drawdown lasts. A drawdown generally is a turning point, and trend-following strategies tend to have the wrong positions initially thereby exacerbating early-stage drawdowns.<sup>2</sup> The longer the drawdown, the more likely the trend-following strategies enter into the right positions. In contrast, DAR4020 typically provides immediate protection as on average it runs at a negative 60/40 beta. An extreme example is the very short ‘Covid’ drawdown Feb-Mar 2020. The 60/40 portfolio lost 12.4% in two months, trend-following returned -1.1%, while DAR4020 gained +5.0%. As trend-following strategies also make a good

<sup>2</sup> Interestingly, Goulding, Harvey, and Mazzoleni (2023a, 2023b) propose an enhanced version of trend-following that effectively blends shorter term and longer-term trends to improve the behavior of trend-following strategies around turning points. Exploring such enhanced trend-following strategies could be a potentially fruitful extension. In the paper we primarily focus on the standard 12-month trend as a representative baseline, while we discuss different trend-following variants in Appendix D.

return in ‘good’ states – states where DAR4020 tends to realize lower returns – we find that both approaches are complementary in providing downside protection. A hedging portfolio based on a combination of 50/50 trend-following and DAR4020 provides robust returns across down- and up-market states, outcompetes the individual strategies, and optimal multi-asset portfolios blend DAR4020 and trend-following with significant weights.

Our paper contributes to the literature in several important ways. First and foremost, we construct a deep sample of 222 years of data over which we evaluate defensive strategies across a variety of drawdown states, to yield the most powerful analyses of defensive strategies to date. As such, we extend the Harvey et al. (2019) sample by up to 188 years. We confirm their positive conclusions on quality equities and trend-following, and their negative conclusions on treasuries and gold, despite these two performing well in their 1985-2018 sample. We further find that the equity strategies low-risk and value based on dividend yield perform well as defensive strategies. Harvey et al. (2019) do not cover low-risk and they conclude negatively on value based on book-to-market ratios. Second, we evaluate and extend the DAR concept in competition with other defensive strategies and show how it outcompetes many other defensive strategies.<sup>3</sup> In addition, we exploit the depth of our two-century sample to examine how defensive strategies behaved across distinct economic scenario and historical regimes – including inflationary and recessionary regimes, pre-Federal Reserve markets, gold-standard eras, major wars, and high-tariff trade-war periods.

The remainder of this paper is constructed as follows. Section 2 presents the deep sample dataset and long-run returns. Section 3 discusses the novel Defensive Absolute Return (DAR) strategy and amendment. Section 4 compares defensive strategies over our deep sample, while Section 5 considers their complementarity, showcases added value to 60/40, and examines robustness across inflationary, recessionary, and expansionary regimes. Section 6 considers other defensive strategies which can only be evaluated over shorter sample periods than our full sample 1800-2021 period, including the equity factors quality, value, and low-risk, and equity index put options, and evaluates protective characteristics against a 100% equity reference portfolio. In the latter case we also analyze Treasuries as defensive strategy from 1800 onwards. In addition, we examine how defensive strategies behaved across distinct historical regimes. Finally, section 7 concludes.

## **2. Data and deep sample investment returns**

We construct a deep sample dataset that spans as much global market drawdowns as possible based on several prominent studies. Our sample covers 222 years of data from 31 December 1799 through 31 December 2021. For constructing the deep sample, we primarily make use of four recent studies that construct good quality datasets over a deep sample. First, Baltussen, Swinkels, and Van Vliet (2021) examine trend, momentum, value, carry, and betting-against-beta factors in equity indices, government bonds, currencies, and commodities from 1800 onwards. Most of these factors have been evaluated from 1981 onwards or even later, and this study finds strong support for these factors for 1800-1980. Second, Baltussen, Van Vliet, and Van Vliet (2023) build long-short equity factors for size, value, momentum, and low-risk from 1866 onwards to provide out-of-sample evidence for these factors prior to the CRSP database start in 1926. Third, Wahal (2019) extends the equity quality factor moving the start from 1964 to 1940. Fourth, Baltussen, Swinkels, van Vliet and van Vliet (2023), construct global asset

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<sup>3</sup> Cavaglia et al. (2022) introduce DAR original but do not put it in competition with other defensive strategies. Harvey et al. (2019) could not cover DAR original as it did not exist at the time of writing their paper. Further, note that Cavaglia et al. (2022) aim to protect against losses of the global 60/40 equities/bonds portfolio, whereas Harvey et al. (2019) look at defensiveness with respect to the S&P 500. In this paper, we analyze defensive properties versus both the 60/40 equities/bonds portfolio and versus the global equities portfolio.

class returns and inflationary data back to 1875. Below we provide a summary of our dataset construction. In Appendix A we outline the data construction more extensively.

Asset class returns. We take global equity and bond returns between 1875 and 2021 from Baltussen, Swinkels, van Vliet and van Vliet (2023), which we splice with data from Baltussen, Swinkels and Van Vliet (2021, henceforth BSV) between 1800 and 1874. All returns are in USD and bond-returns are currency-hedged. The 60/40 portfolio is constructed as 60% market cap weighted equities (total return, USD) and 40% global GDP-weighted bonds (hedged USD), GDP data is available from April 1871, global bonds are equally weighted before. Further, we also collect gold prices and returns from Global Financial Data (GFD) and Bloomberg.

Factor returns. Factor returns come from various sources. First, BSV provide global factor returns across asset classes. For market-level indices or futures in equities, bonds, currencies (FX), and commodities, BSV construct Time-series momentum (Trend, see Moskowitz, Ooi, and Pedersen, 2012); Cross-sectional momentum (Momentum) and Value (both see Asness, Moskowitz, and Pedersen, 2013); Carry (see Koijen, Moskowitz, Pedersen, and Vrugt, 2018); Seasonal, and Betting-Against-Beta (BAB, see Frazzini and Pedersen, 2014) factor strategies. In this dataset the most recent historical data are from Bloomberg, Datastream and OECD and before that from GFD.

Second, for U.S. equities we use CRSP from 1926 onwards, and the novel dataset from 1866-1926 from Baltussen, Van Vliet and Van Vliet (2023). The older data come from GFD and the Commercial and Financial Chronicle (CFC; the source used to build the CRSP sample as of 1926). For U.S. equities Baltussen, Van Vliet, and Van Vliet (2023) construct Value (High minus Low based on dividend yield); Size (Small minus Big based on firm size, see Fama and French, 1992, 1993); Momentum (Winners minus Losers akin to Jegadeesh and Titman, 1993); and BAB based on volatility (see Blitz, Van Vliet, and Baltussen, 2020).<sup>4</sup> Third, for the longest sample on quality in U.S. equities we use data from Wahal (2019), who extends history to 1940 for investment and profitability, see also Fama and French (2012) and Novy-Marx (2013). Quality is constructed as an equal-weighted average of investment and profitability. For more specific details on each dataset, we refer to the aforementioned studies.

The equal volatility-weighted combination of the Time-series momentum factors from BSV in equities, bonds, currencies, and commodities (i.e., Trend Bonds, Trend Commodities, Trend Equity Index and Trend FX) forms the main trend-following strategy in the paper. Note that this definition is based on trend-following signals based on the past 12 months return skipping the most recent month to control for potential spurious short-term autocorrelation due to different data quality in the deeper sample, see BSV for more detail. In Appendix D we confirm robustness to alternative trend-following windows.

Before we continue with describing strategy returns a discussion on data quality and historical markets is in order. Deep historical data tends to be of lesser quality compared to the more recent data, as digital archives and strong requirements on data processes did not exist. Instead, data was maintained typically by exchanges, statistical agencies, newspapers, and investor annuals, often in manual writing. Potential data quality issues that could be at work include misprints and other measurement errors, but also the use of old data, the use of time-averaged prices over a month, and the timing of dividends or coupons sometimes being unknown but assigned at quarter or year ends. Moreover, asset and factor strategies were typically composed of fewer markets, especially during the 19<sup>th</sup> century. Lower data quality could potentially influence asset and factor returns by introducing random measurement errors, increases risk, spurious correlations, or creating systematic biases correlated with factor

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<sup>4</sup> Baltussen, Van Vliet, and Van Vliet (2023) construct Value based on dividend yield as in the 19<sup>th</sup> century dividends were widespread, strongly associated with earnings, and seen as the common valuation technique for stocks.

premium. The aforementioned studies address these issues by paying significant effort to constructing high-quality datasets and factor series. They for example cross-validate data across sources, remove factor series during periods with too few high-quality asset returns, and apply various conservative data-quality filtering screens that reduce issues related to liquidity, tradability, and other potential data inaccuracies. While these screens enhance data quality and reduce biases, they are unlikely to fully eliminate all potential impacts. Such concerns are particularly relevant for data from the 19<sup>th</sup> and early 20<sup>th</sup> centuries but are less applicable to the more recent subsamples also analyzed in this study. Nevertheless, there is no clear reason to expect these remaining biases to significantly affect the defensive properties, as the aforementioned studies show factor returns and dynamics are generally robust to data quality considerations.

For the 25 factors, Table 1 presents the average return per month, the full-sample correlation with the 60/40 equity/bond returns, the start date, and the number of observations. Rows on factor returns are sorted by correlation with the 60/40 portfolio. Note that Harvey et al. (2019) start in 1985 and end in 2018 and we add up to 188 years of data. Cavaglia et al. (2022) start in 1926 and use a partially different factor set, and hence we add up to 60 years of extra data on U.S. equity factors and up to 126 years of extra data on the other factors. Equities, bonds, and the 60/40 portfolio offer attractive long-term returns. Global equities returned 8.1% per annum, global (currency-hedged) bonds 5.2%, and the 60/40 multi-asset portfolio 7.0%. Most individual factors offer attractive returns and close to zero correlation with the 60/40 portfolio. Full sample Value U.S. Equities has the most negative correlation with 60/40. Value is based on dividend yield, hence the top portfolio has a bias towards large stocks with low risk and solid profits which will tend to do relatively well in crisis periods. These are full sample correlations and through time correlations fluctuate around these means.

**Table 1: Factor summary statistics**

| Factor                 | Mean (%) | 60/40 correlation | Start date | Nr. of obs. |
|------------------------|----------|-------------------|------------|-------------|
| Value U.S. Equities    | 1.12%    | -0.32             | Jan-1866   | 1,872       |
| Quality U.S. Equities  | 7.22%**  | -0.21             | Jul-1940   | 978         |
| Momentum U.S. Equities | 6.33%**  | -0.14             | Jan-1866   | 1,872       |
| Trend Bonds            | 6.36%**  | -0.10             | Jan-1800   | 2,664       |
| Trend Commodities      | 6.15%**  | -0.05             | Mar-1878   | 1,726       |
| Momentum Equity Index  | 5.72%**  | -0.04             | Jan-1800   | 2,664       |
| Value FX               | 1.15%    | -0.04             | Mar-1871   | 1,810       |
| Value Commodities      | 2.60%**  | -0.02             | Feb-1878   | 1,727       |
| Momentum Bonds         | 2.47%**  | -0.01             | Oct-1800   | 2,655       |
| Value Equity Index     | 2.00%**  | 0.00              | Jan-1800   | 2,664       |
| Value Bonds            | 2.45%*   | 0.00              | Mar-1871   | 1,810       |
| Carry Equity Index     | 6.03%**  | 0.00              | Jan-1800   | 2,664       |
| Carry FX               | 5.52%**  | 0.01              | Mar-1800   | 2,662       |
| Momentum FX            | 3.14%**  | 0.01              | Mar-1800   | 2,662       |
| Trend Equity Index     | 8.97%**  | 0.02              | Jan-1800   | 2,664       |
| Trend FX               | 4.78%**  | 0.02              | Mar-1800   | 2,662       |
| Momentum Commodities   | 5.79%**  | 0.02              | Mar-1878   | 1,726       |
| BAB Bonds              | 0.44%    | 0.02              | Oct-1800   | 2,655       |
| Carry Commodities      | 2.41%*   | 0.03              | Feb-1878   | 1,727       |
| Carry Bonds            | 5.79%**  | 0.06              | Feb-1800   | 2,663       |
| BAB FX                 | 0.35%    | 0.09              | Apr-1801   | 2,649       |
| BAB U.S. Equities      | 5.81%**  | 0.09              | Jan-1866   | 1,872       |
| BAB Commodities        | 0.82%    | 0.10              | Apr-1878   | 1,725       |
| Size U.S. Equities     | 1.69%    | 0.12              | Jan-1866   | 1,872       |
| BAB Equity Index       | 3.04%**  | 0.16              | Jan-1800   | 2,664       |

Note: The data covers monthly returns from January 1800 to December 2021. Each factor uses a 10-year rolling window to target an ex-ante volatility of 10% per annum. Annualized means and the correlation are based on the full-sample monthly returns. The label U.S. Equities is used for the individual stock-level factors, while Equity Index is used for the factors based on international equity indices/futures. \* and \*\* denotes significance at the 5% and 1% significance level, respectively.

### 3. Defensive Absolute Return

In this section, we first discuss the novel Defensive Absolute Return (DAR original) strategy proposed in Cavaglia et al. (2022) and its enhanced versions.

As mentioned in the introduction, the DAR portfolio is constructed by sorting on past correlation with a 60/40 index. Cavaglia et al. (2022) use rolling correlations of each factor with a 60/40 index using a 60-month lookback window. On a monthly basis these correlations are used as a sorting variable. A hedging portfolio is created that is long the 33% factors that are most negatively correlated to the 60/40 and short the 33% factors that are most positively correlated to the 60/40. Factors are equally weighted both on the long and the short side. The middle 33% of factors is discarded. For our factor set – which include all factors shown in Table 1 as of their starting date (note that this also includes the trend factors) – the weighting scheme translates in eight factors in the long portfolio and eight factors in the short portfolio. As mentioned in the introduction, we will refer to this solution as DAR original. We are the first to put the concept into competition with other protective strategies, perform a deep out of sample test and use a partly deviating factor set from the original paper.

Having the same number of long and short factors on average leads to a zero long-term performance of DAR original, under the assumption that long term factor performance is equal across factors. Hence the logic in Cavaglia et al. (2022) that DAR original provides protection for the 60/40 portfolio at zero return costs. However, a full sample average performance of zero with a significant volatility around that, also implies there will be drawdown periods. Especially in prolonged 60/40 bull markets such drawdown periods might be too extensive to bear. Further, Harvey et al. (2019) make the case for only considering protective strategies that have long-run positive gross returns. These returns can be used to pay for trading costs and support the overall protective profile. Considering the strong long-run evidence on factor premiums (see Table 1), we next investigate alternatives by going long more factors than short factors. By having a net positive factor exposure, a net positive performance is achieved.

In Table 2 we reconsider the construction of the top and bottom portfolio (top or bottom terciles in Cavaglia et al.). We cover four sample starting dates for robustness purposes. Starting in 1926/1 matches the starting date of the original sample period in Cavaglia et al. (2022) and allows for a comparison with their results. The post 1800/1 sample is the deepest period available with many factors included and serves as our main sample given the prominence of most 60/40 drawdowns and statistical power. This is the longest sample that can be constructed to examine the downside strategies we consider. Further, we consider two additional subsamples for robustness purposes. As of 1878/4 commodity factors and the equity selection factors Value, Momentum, Size and BAB enter the universe. As of 1940/7 quality is available and all factors can be included.

**Table 2: Varying the long and short leg of DAR across samples**

| Long/short | Correlation (60/40, DAR) |          |          |          | Volatility (60/40+DAR) |          |          |          |
|------------|--------------------------|----------|----------|----------|------------------------|----------|----------|----------|
|            | 1/3, 1/3                 | 0.4, 0.2 | 0.5, 0.1 | 0.6, 0.0 | 1/3, 1/3               | 0.4, 0.2 | 0.5, 0.1 | 0.6, 0.0 |
| 1926/1     | -0.39                    | -0.37*   | -0.31**  | -0.19**  | 8.8%                   | 8.9%     | 9.2%     | 9.7%     |
| 1800/1     | -0.28                    | -0.28    | -0.24**  | -0.17**  | 7.9%                   | 7.9%     | 8.1%     | 8.5%     |
| 1878/4     | -0.34                    | -0.33    | -0.27**  | -0.19**  | 8.0%                   | 8.1%     | 8.3%     | 8.7%     |
| 1940/7     | -0.43                    | -0.42*   | -0.34**  | -0.20**  | 8.5%                   | 8.6%     | 9.0%     | 9.6%     |

Note: Ex-post correlations of Defensive Absolute Return variants with 60/40. The table depicts how many factors end up in the long and short legs (1/3 each, 40% long and 20% short, 50% long and 10% short, and 60% long and no shorting). Column 1 shows the starting point of the sample, with 2021/12 the end date in all four cases. The volatility columns show the volatility of the 60/40 portfolio with the DAR portfolio included as an overlay. For correlations significance statistics are added, measured as deviation from the base case '1/3, 1.3'. \* and \*\* denotes significance at the 5% and 1% significance level, respectively.

We find that the ex-post correlations with the global 60/40 equity/bond portfolio are consistently negative. Moreover, the hedging quality, as measured by the ex-post correlation with the 60/40, remains largely intact when moving from 33% long and 33% short (DAR original, columns ‘1/3, 1/3’) to 40% long and 20% short (DAR4020, columns ‘0.4, 0.2’), which has ten factors in the long portfolio and five factors in the short portfolio when all factors are available. Table 2 shows that for the longest sample, 1800-2021 (Table 2 row ‘1800/1’), the ex-post correlation remains the same at  $-0.28$ . In the 2 recent samples we find significant statistics at the 95% level of confidence of less negative correlation. When further increasing the long leg at the cost of the short leg, defensive properties start to diminish as correlations start to increase, from  $-0.28$  to  $-0.24$ , and to  $-0.17$  for the 60% long with 0% shorting variant, all significant at the 99% confidence level. The same conclusion is found when considering the joined volatility of the 60/40 portfolio with the DAR overlay included. Full sample volatility of the 60/40 portfolio stands at 7.7%. For both the ‘1/3, 1/3’ (DAR original) and ‘0.4, 0.2’ (DAR4020) variants, volatility marginally increases to 7.9%. When moving beyond, the overall risk profile increases further. Considering the deepest sample, moving to ‘0.5, 0.1’ increases volatility by 0.2% and correlation by 0.04 versus DAR original, and for ‘0.6, 0.0’ volatility increases by 0.6% and correlation by 0.11. The story behind the risk figures is that the more we move to a purely long factor portfolio, the more we move from a put-like pay-off to a more ‘all weather’ alternative risk premia solution. Defensive properties deteriorate, while the overall return profile increases. The shorter sample periods provide a similar picture. Our results indicate we can move to Defensive Absolute Return with long the 40% factors with the most negative correlation with 60/40 and short the 20% factors with the most positive correlation with 60/40 without materially harming defensive properties as measured by negative correlation and overall risk by volatility. As such, we continue with this ‘0.4, 0.2’ version (DAR4020) and assess performance numbers.

**Table 3: Defensive Absolute Return ‘original’ and ‘4020’ stand-alone performance**

|                    | Return p.a. | Stdev p.a. | IR   | Corr 60/40 | Beta 60/40 | 60/40 up  | 60/40 down |
|--------------------|-------------|------------|------|------------|------------|-----------|------------|
| Panel A: 1800-2021 |             |            |      |            |            |           |            |
| DAR original       | 0.8%        | 5.0%       | 0.16 | $-0.28$    | $-0.18$    | $-0.15\%$ | 0.49%      |
| DAR4020            | 2.3%        | 5.0%       | 0.46 | $-0.28$    | $-0.18$    | $-0.03\%$ | 0.61%      |
| Delta              | 1.5%**      |            |      | 0.00       |            | 0.12%**   | 0.12%**    |
| Panel B: 1926-2020 |             |            |      |            |            |           |            |
| DAR original       | 0.8%        | 5.0%       | 0.16 | $-0.39$    | $-0.20$    | $-0.25\%$ | 0.70%      |
| DAR4020            | 2.8%        | 5.0%       | 0.55 | $-0.37$    | $-0.19$    | $-0.07\%$ | 0.81%      |
| Delta              | 2.0%**      |            |      | 0.02*      |            | 0.19%**   | 0.11%**    |

Note: Ranking factors on ex-ante 60-month lookback correlations with 60/40, DAR original is long the top tercile most negative correlations and short the bottom tercile most positive correlations. Ranking factors on ex-ante 60-month lookback correlations with 60/40, DAR4020 is long the top 40% and short the bottom 20%. Shown are the stand-alone annualized performances and volatilities in excess of cash, the ex-post realized correlations and betas with 60/40, and the average monthly performance conditional on 60/40 being positive (‘up’) or negative (‘down’). Panel A includes the full sample 1800-2021. Panel B aligns to the Cavaglia et al. (2022) sample at 1926-2020/4. Each protective strategy is scaled to a 5% volatility based on its ex-post realized volatility over the sample period to make them comparable. \* and \*\* denotes significance at the 5% and 1% significance level, respectively.

Table 3 summarizes the key characteristics of the original solution, DAR original, versus the amended DAR4020 solution. Note that here, and in most other cases below, we use the full- (or sub-)sample realized volatility to retroactively scale strategy returns to 5% volatility such that statistics are presented as comparative metrics across defensive strategies. This means reported returns are normalized outcomes, not necessarily actual portfolio returns an investor would have gotten with a more dynamic, ex ante leverage adjustment. We focus on the deepest sample in panel A, as well as the original sample by Cavaglia et al. (2022) in Panel B. Panel A reveals that both variants provide on average protection when the 60/40 portfolio has negative

returns ('60/40 down'), resulting in the negative correlations. By earning from on average a net long exposure to factor premia, DAR4020 has a positive stand-alone performance of 2.3% per annum for 5% volatility, a high average return when 60/40 is down, and small losses when 60/40 is up. Compared to DAR original, the amended DAR4020 version performs significantly better, with higher returns overall and significantly higher returns in both the up and down states. It realizes these improved returns without materially harming defensive properties, as indicated by negative correlations and betas with respect to the 60/40 index and the high returns in 60/40 down states.

Panel B shows that these results are comparable over the sample period used Cavaglia et al. (2022), with DAR4020 offering better defensive properties and higher unconditional average returns. Moreover, comparing our results to Cavaglia et al. reveals strong defensive properties across both studies.

DAR4020 is dynamic regarding which factors populate the long and short leg because the rolling correlations change over time. Some factors, however, can be found almost exclusively in one of these two legs. Given the full-sample statistics in Table 1 it should not come as a surprise that the long leg – using factors that have a negative correlation with 60/40 – very often is populated with value, quality, and momentum U.S. Equities.<sup>5</sup> The short leg often includes Size Equities and BAB strategies. BAB strategies are long low beta assets and short high beta assets, where the long leg is leveraged up to the same risk profile as the short leg. In times of crisis, correlations move up, and these strategies tend to be over-leveraged.

Note that throughout this paper we do not consider the impact of transaction costs and, hence, the defensive strategies we evaluate are not optimized for it. An important implementation question concerns the optimal tradeoff between transaction costs and defensive properties. Given that trading on all signals requires frequent rebalancing, and strategies are meant as addition to traditional portfolios, there may be extensive netting across portfolio positions. Moreover, there are other manners that limit turnover or trading cost, for example using defensive strategies in the timing of already planned trades, buffering on correlation ranking, optimizing rebalance frequencies, the universe choice, or factor set against turnover. We leave a thorough assessment of the impact of costs and frictions, or the examination of the optimal implementation for future research.

Additionally, it is important to note that historical investability differs from today's standards. Although investment frictions are generally assumed to have been higher in the 19<sup>th</sup> century compared to the 21<sup>st</sup> century, evidence suggests that trading in historical markets was neither impossible nor excessively costly. Asset markets during the 19<sup>th</sup> century shared many similarities with modern markets, including active and liquid derivatives markets with short-selling capabilities, established margin trading systems, market-making and arbitrage practices, and centralized exchanges equipped with clearing mechanisms. Stocks, bonds, currencies, and commodities traded internationally, and often at limited transaction costs. Active trading, including short-selling, was common in the U.S. stock market, and transaction costs in the 19<sup>th</sup> century were comparable to those in the 20<sup>th</sup> century (e.g., Jones, 2002, Brown et al., 2008). For a more in-depth discussion of historical investment frictions and the history of financial markets, we refer to the Online Appendices of BSV, and Baltussen, Van Vliet and Van Vliet (2023), as well as Neal (1985, 1987), Harrison (1998), Poitras (2009), Accominotti and Chambers (2016), Chambers and Dimson (2016), and Koudijs (2016), among others. That said, we like to emphasize that our deep-sample results represent historical simulations under modern definitions, and real-world historical frictions, such as illiquidity, trading costs, limited

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<sup>5</sup> In section 6 we consider several individual factor strategies as defensive strategies across different samples, showing DAR4020 generally performs superior.

availability of derivatives, or capital restrictions, would have influenced strategy results historically.

#### **4. Tail risk hedging: Contrasting defensive strategies in tail scenarios starting in 1800**

We next contrast several defensive strategies – including DAR4020 – over a sample spanning 222 years and 8 drawdowns on the 60/40 global equity/bond portfolio exceeding 20%. As such we provide a large out-of-sample test of defensive strategies, with previous studies focusing on drawdowns and recessions starting in 1985 and which tend to be restricted to the U.S. We examine defensive properties from various angles: co-movement with 60/40 performance and performance conditional upon the global 60/40 portfolio being down or up in section 4.1, performance in 60/40 drawdown periods in section 4.2, and performance across 60/40 distribution quantiles in section 4.3. Further, in section 4.4 we discuss robustness across in-sample and out-of-sample periods. Our examination considers the defensive strategies that we can evaluate over most part of our sample: trend-following, gold, and DAR4020. In section 6 we consider the low-risk, value, and quality factors in equities – available since 1866 or 1940, respectively – put option strategies – available since 1986 – and downside protection for equity-only portfolios from 1800, where we also include government bonds as defensive strategy.

##### **4.1 Key performance characteristics of protective strategies**

Table 4 shows the characteristics of the hedging strategies available from 1800 onwards. Gold is clearly the odd one out, with a positive correlation with 60/40 and on average negative performance when 60/40 is down. Even if one would doubt the role of gold from 1800 to 1939, results in section 6.3 show that gold also has the weakest defensive properties across our evaluated strategies for 1940-2021.

DAR4020 shows to be the strongest hedge with a correlation of  $-0.28$  and an average performance of  $0.61\%$  in 60/40 down months. Performance-wise, trend-following is the strongest with an IR of  $0.97$  while it also returns well during 60/40 down months. Related, Harvey et al. (2019) show that trend-following strategies are one of the most effective defensive strategies during crisis periods, and Hurst, Ooi, and Pedersen (2017) show that trend-following strategies perform especially well across various recessions and crisis periods since 1880. Note that DAR4020 has on average negative returns in positive 60/40 months. Hence in long bull markets the strategy will on average have negative returns, which will test the patience of the investor using it as a protective strategy.

**Table 4: Performance characteristics protective strategies 1800-2021**

|                 | Return<br>p.a. | Stdev<br>p.a. | IR    | AR   | Corr<br>60/40 | Beta<br>60/40 | 60/40<br>up | 60/40<br>down |
|-----------------|----------------|---------------|-------|------|---------------|---------------|-------------|---------------|
| 60/40           | 7.0%           | 7.7%          |       |      | 1             | 1             | 1.64%       | -1.49%        |
| DAR4020         | 2.3%           | 5.0%          | 0.46  | 0.74 | -0.28         | -0.18         | -0.03%      | 0.61%         |
| Gold            | -0.7%          | 5.0%          | -0.14 | 0.09 | 0.17          | 0.11          | 0.06%       | -0.30%        |
| Trend-following | 4.8%           | 5.0%          | 0.97  | 1.01 | -0.05         | -0.03         | 0.41%       | 0.38%         |

Note: Ranking factors on ex-ante 60-month lookback correlations with 60/40, DAR4020 is long the top 40% and short the bottom 20%. The first four columns show the stand-alone performance, including annualized return, annualized volatility, the ratio of return and volatility per annum (IR) and the Appraisal Ratio (AR) by regressing the returns on 60/40 to correct for the beta. The other columns show the ex-post realized correlation and beta with 60/40, and the average monthly performance conditional 60/40 being positive ('up') or negative ('down'). Each protective strategy return is scaled to a 5% volatility based on its ex-post realized volatility over the sample period to make them comparable, and all returns are in excess of cash.

## 4.2 Tail-risk properties of protective strategies

Correlations and average returns in down months might misrepresent important dimensions of risk, especially when return distribution are skewed or have fat tails. An alternative is to look at the behavior during drawdowns. Van Hemert et. al (2020), for example, study theoretical properties of drawdowns, and argue for the practitioner relevance of drawdown management. In fact, asset managers and fiduciaries routinely use the drawdown statistic for fund allocation and redemption decisions. For tail risk, we next look at periods where 60/40 loses more than 20%. We consider a high watermark approach; look at the cumulative performance of the 60/40 through time and at each month look how far it is below the maximum reached before that, shown earlier in Figure 1. When we reach a trough, the difference with the previous maximum is the drawdown. Table 5 shows for 1800-2021 the eight drawdowns when 60/40 lost 20% or more. The 60/40 equity/bond portfolio on average loses 37.6% in these eight cases. Both DAR4020 and trend-following provide protection, with average *positive* returns of 16.6% and 19.3% respectively, with DAR4020 more reliable at drawdown returns ranging from 4.0% up to 36.9% (difference 33.0%), versus trend-following at -5.1% up to 35.2% (difference 40.3%). In contrast, Gold provides no protection on average. Note that for interpretability and comparability, each defensive strategy is scaled to an ex-post realized annualized volatility of 5%, whereas the 60/40 strategy remains unscaled, exhibiting a full-sample realized volatility of 7.7%.

In contrast to common intuition, we find that gold does not deliver on average as defensive strategy. This aligns with the findings of Harvey et al. (2019) for the 1985 – 2018 period, who find relatively poor defensive properties of gold while focusing on the S&P 500 drawdowns. Similarly, Erb and Harvey (2013) find that over an extended historical sample gold is an unreliable equity hedge. Instead, Harvey et al. (2019) conclude that the best downside protection is provided by trend-following strategies and the quality factor in equities. We confirm their findings over a deeper sample for trend-following, adding the DAR4020 approach as effective defensive strategy. In sections 6.3 and 6.4 we will examine the quality factor in equities, as this factor is only available over a more limited sample period as of 1940.

**Table 5: 60/40 drawdowns of 20% or more 1800-2021**

| Start DD | Worst DD | 60/40  | DAR4020 | Gold   | Trend-following |
|----------|----------|--------|---------|--------|-----------------|
| 1846/7   | 1848/3   | -24.9% | 10.5%   | -4.0%  | 35.2%           |
| 1864/8   | 1865/5   | -21.2% | 4.0%    | -27.0% | -5.1%           |
| 1918/10  | 1920/12  | -71.2% | 7.7%    | -5.2%  | 20.2%           |
| 1929/2   | 1932/6   | -45.0% | 16.3%   | -3.6%  | 31.5%           |
| 1944/11  | 1948/7   | -36.1% | 17.9%   | 0.2%   | 28.2%           |
| 1973/7   | 1974/9   | -32.7% | 31.8%   | 9.2%   | 20.0%           |
| 2000/4   | 2002/9   | -30.9% | 36.9%   | 2.9%   | 21.2%           |
| 2007/11  | 2009/2   | -39.2% | 7.4%    | 8.1%   | 3.3%            |
| Avg. all |          | -37.6% | 16.6%   | -2.4%  | 19.3%           |

Note: We look at periods where 60/40 lost more than 20%. In total there are eight such periods, indicated by the first two columns. The total return loss of the 60/40 is shown in the third column. The other columns show the cumulative returns in excess of cash of DAR4020, gold, and trend-following over the 60/40 drawdown windows. The 60/40 returns represent unscaled returns, while the returns on all presented defensive strategies are scaled to 5% volatility per annum based on its ex-post realized volatility over the sample period to make them comparable.

## 4.3 Additional tail risk measures

Next, we consider yet another tail risk measure by examining return behavior in various quantiles of the 60/40 return distribution. For reasons of brevity, we present the results in Appendix B. From Table B1 we see again that DAR4020 provides the best protection in poor months for the 60/40 irrespective of the quantile and the larger the loss, the larger the positive performance from DAR4020. Further, Figure B1 shows another way to illustrate the hedging

ability by sorting the 60/40 and all protective strategy returns on the size of the 60/40 returns, from the most negative 60/40 month to the most positive 60/40 month. Subsequently we compute the cumulative returns of the sorted monthly returns. Overall, DAR4020 has the best average returns in the roughly 700 months with the worst 60/40 returns, outperforming trend-following.

#### **4.4 Robustness to in- and out-of-sample periods**

Next, we explicitly consider the robustness of our results across in-sample and out-of-sample periods and present the results in Appendix C. We consider the periods used by either Cavaglia et al. (2022) or Harvey et al. (2019) as in-sample periods, and the periods between 1800 and the start of their samples as the out-of-sample periods. Table C1 summarizes the results, revealing that the defensive property of DAR4020 is robust to the sample choice, with strong returns across drawdowns, the 10% worst 60/40 months, as well as 60/40 down months. Trend-following is also relatively robust as a performance driver in both up and down states across samples, with only the recent Harvey et al. (2019) sample in panel C providing weaker defensive performance. Gold provides weaker and inconsistent defensive properties across samples and underperforms both DAR4020 and trend-following, so we do not include it in the next section.

### **5. The complementary nature of DAR4020 and trend-following**

Having established the defensive performance of especially DAR4020 and trend-following, we next examine the difference (section 5.1) and complementary nature (section 5.2) of DAR4020 and trend-following. In addition, we exploit the depth of our two-century sample to examine how defensive strategies – including the combination of DAR4020 and trend-following – performs across various economic and market scenarios that differ from downside states based on historical asset class returns.

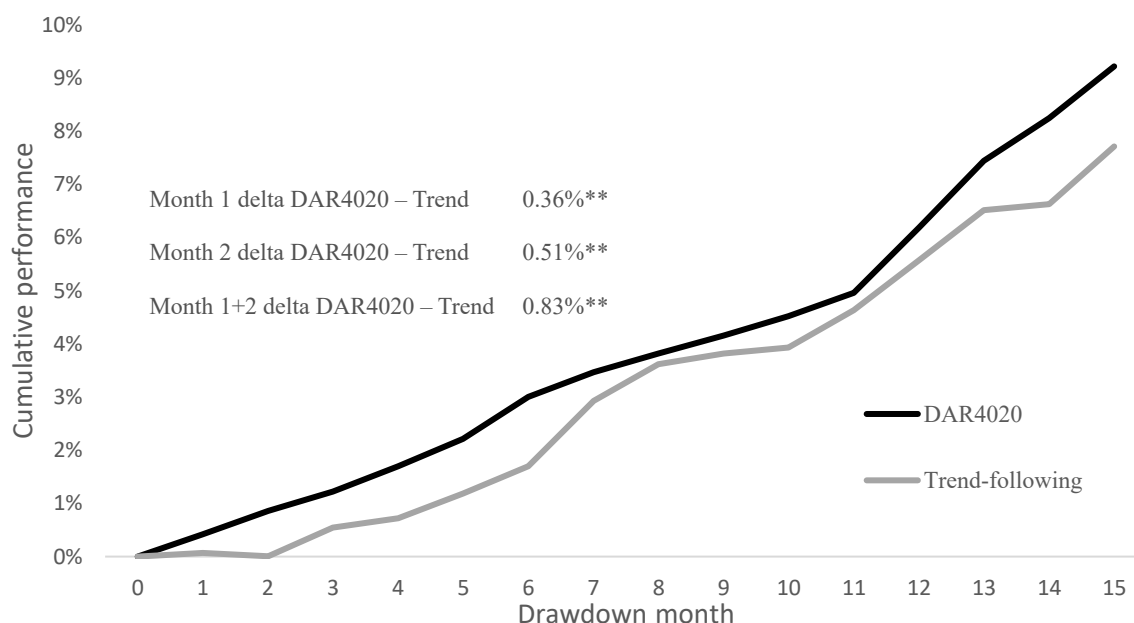
#### **5.1 Understanding the difference between DAR4020 and trend-following**

Above we have shown that DAR4020 and trend-following are effective downside protection strategies over our deep sample, but also that they behave differently through time. We further examine the effectiveness of DAR4020 versus trend-following strategies in offering downside protection. What about the ‘sharp’ drawdowns where the put has a potential advantage over trend-following (see for example Ilmanen et al., 2021)? In Table 5 we saw that the 1864/8 – 1865/5 drawdown has the shortest length at 10 months, and it is the only one where trend-following has a negative performance. Sharp drawdowns last by definition shorter. Evaluating all 60/40 drawdowns, we next also examine the average performance over all 1<sup>st</sup> months, all 2<sup>nd</sup> months, etcetera. This result is shown in Figure 3.

The performance difference between DAR4020 and trend-following is significantly positive over the first two months, both standalone (0.36%, 0.51%) and joined (0.83%). From month three onwards, performance differences are no longer significant. As DAR4020 is running a negative beta it on average provides immediate protection at the start of drawdowns, in contrast to trend-following where the trend strategies need time to get into the right position. The trend-following return is on average zero at the start of drawdowns and hence does not provide immediate protection. Only in the third month does average performance become positive. Trend-following can initially make drawdowns worse. In case of sharp and shorter drawdowns, like the Covid related recession in 2020, trend-following realizes negative returns during a downward market. In contrast, DAR4020 provides on average immediate protection, similar to a put option. It further becomes apparent that trend-following does perform well over the whole drawdown, after getting in the right position, so that the slope of the cumulative

return charts of two strategies becomes similar. More details on severe drawdowns and relative performance across strategies may be found in Table 5.

**Figure 3: Cumulative average performance from the start of drawdowns**



Note: We consider drawdown months of the global portfolio investing 60% in equities and 40% in bonds. For each month we calculate the average cumulative performance (in excess of cash) of DAR4020 and trend-following over all drawdown episodes. A 60/40 drawdown period is completed when a new high watermark is reached. In total there are 281 drawdown periods from 1800 to 2021 where we can evaluate DAR4020 and trend-following performance. Of course, the further out the less drawdowns are in the sample. There are only 8 drawdowns lasting longer than 15 months, and most of these drawdowns are included in Table 5. Above the cumulative performance lines we show the average return gap over all drawdowns between DAR4020 and trend-following for the first two months. \*\* denotes significance at the 1% significance level. All other differences from drawdown month 3 onwards are not statistically significant.

## 5.2 Combining DAR4020 and trend-following with 60/40

We propose to combine protective strategies, to benefit from their relative strengths and to diversify away the weaknesses. For the period 1800-2021 DAR4020 and trend-following have a correlation of 0.29, an indication of potential diversification benefits from combining. In particular, we have seen that DAR4020 provides immediate protection as it runs at a negative beta but will lose in strong bull markets. And trend-following does on average provide protection but not immediately at the start of drawdowns as it first needs to get into the right positions and therefore struggles in short and steep drawdowns; but it may help offset DAR4020 losses in long bull markets.

We combine DAR4020 and trend-following by assigning them a weight and ensuring that the resulting portfolio has a volatility of 5% to make it comparable to the individual protective strategies. Table 6 shows the results. An arbitrary 50-50 combination of DAR4020 and trend-following provides more protection in the eight drawdowns of 60/40 larger than 20% shown (column '>20% DD') and all 60/40 drawdowns in excess of 2% (column '>2% DD') than using only DAR4020 or only trend-following. Compared to just using DAR4020 most statistics improve while the strength of DAR4020 in '60/40 down' is undiminished. Compared to just using trend-following all defensive characteristics improve, at the cost of less return in positive 60/40 months, leading to a somewhat lower average return per annum. Beta-adjusted, the Appraisal Ratio improves from 1.01 to 1.10.

**Table 6: Combining protective strategies**

|                 | Return<br>p.a. | Stdev<br>p.a. | IR   | AR   | >20%<br>DD | >2%<br>DD | Worst<br>10% | 60/40<br>up | 60/40<br>down |
|-----------------|----------------|---------------|------|------|------------|-----------|--------------|-------------|---------------|
| DAR4020         | 2.3%           | 5.0%          | 0.46 | 0.74 | 16.6%      | 2.8%      | 1.0%         | −0.03%      | 0.61%         |
| Trend-following | 4.8%           | 5.0%          | 0.97 | 1.01 | 19.3%      | 2.1%      | 0.5%         | 0.41%       | 0.38%         |
| 50-50           | 4.4%           | 5.0%          | 0.89 | 1.10 | 22.5%      | 3.1%      | 0.9%         | 0.24%       | 0.62%         |

Note: We look at combining protective strategies. Each protective strategy return is scaled to 5% volatility based on its ex-post realized volatility over the sample period to make them comparable to each other, including the 50-50 combination. The Information Ratio (IR) is the ratio for the annualized return and volatility (Stdev). The Appraisal Ratio (AR) adjusts the IR for the 60/40 beta exposure. We look at cumulative average performance over all drawdowns in excess of 20% ('>20% DD') or 2% ('>2% DD'), the percentile of worst 10% returns ('Worst 10%'), and the average performance in months that 60/40 has a positive return ('60/40 up') or a negative return ('60/40 down'), with the latter three presenting average monthly returns. The sample period is 1800-2021 and all returns are in excess of cash.

Note that so far and below Trend-following is based on a 12 months lookback window. As a robustness check in Appendix D we look at shorter or blended (i.e., shorter and longer) lookback windows. Table D1 shows that the slower trend variant with a 12 months lookback performs best, and is superior over shorter trend lookback windows. Further, defensive properties of the faster trend variants tend to be worse. Hence using the 12 months lookback window for Trend-following benefits Trend-following.

Next, we examine an investor looking to add defensive strategies to the 60/40 portfolio. The results are presented in Table 7. In Panel A we naively add DAR4020, trend-following, or 50/50 DAR4020/Trend (as covered in Table 6) to the 60/40 portfolio. In Panel B we let an optimizer decide on the optimal allocation to DAR4020 and trend-following, either by looking at minimizing risk or by maximizing return divided by risk. In the latter case we either allow volatility to exceed that of 60/40, or we require the volatility of the combination of 60/40 with DAR4020 and Trend to be the same as that of 60/40 stand-alone.<sup>6</sup>

In Panel A, adding DAR4020 slightly increase portfolio volatility to 7.9%, while adding trend-following increases the portfolio risk to 9.0%. This can be explained by DAR4020 running at a correlation of −0.28 compared to −0.05 for trend-following. If the objective is to add a protective strategy to 60/40 while keeping long-term portfolio volatility close to that of 60/40, then DAR4020 is preferable over trend-following.

In down markets, adding DAR4020 to the 60/40 portfolio improves returns significantly more than adding trend-following. However, in 60/40 up markets trend-following provides benefits, as on average it protects in 60/40 drawdowns, but also has an expected positive return in good states, significantly outperforming DAR4020. In contrast DAR4020 makes most returns in the bad states, which is why it is such a good defensive strategy. But on average it has small negative returns in good states. This will test the resolve of users in strong bull markets. It is therefore interesting to combine DAR4020 and trend-following.

Adding 50% DAR4020 and 50% trend-following to 60/40 increases portfolio volatility from 7.7% to 8.3%, 0.7% less than adding only trend. Hence this 50-50 combination provides a better hedge than using one of the solutions alone. Returns in down states materially improve, as do worse than 2% drawdowns, worst 10% of observations as well as average performance in negative 60/40 months. This does come at the cost of significantly less overall return coming from the 60/40 up states, but these extra returns do come at high relative risk levels, considering 0.4% in extra return costs an additional 0.7% in volatility. In terms of 60/40 drawdowns in

<sup>6</sup> In recent work by Man AHL named '*Honey, I Shrunk the Trend-Following*' (2024) they maximize the Sharpe ratio while allocating between 60/40 and trend-following, finding an optimal weight of 60% to 60/40 and 40% to trend-following. In contrast, we keep the 60/40 allocation fixed at 100% and allow the addition of other strategies through leverage.

excess of 20%, 60/40 on average loses 37.6%. This reduces to 21.1% when adding DAR4020, 18.3% when adding trend-following, and 15.1% with a 50/50 combination of DAR4020 and trend-following. 60/40 combined with equally weighted DAR4020 and trend-following as hedging portfolio realizes a competitive IR at 1.37 ('60/40+50/50'). Versus 60/40 standalone, drawdowns are significantly reduced, and returns are significantly higher across key metrics.

**Table 7: Combining 60/40 with protective strategies**

|                                                             | Return<br>p.a. | Stdev<br>p.a. | IR   | >20%<br>DD | >2%<br>DD | Worst<br>10% | 60/40<br>Up | 60/40<br>down |
|-------------------------------------------------------------|----------------|---------------|------|------------|-----------|--------------|-------------|---------------|
| <b>Panel A: Heuristic additions</b>                         |                |               |      |            |           |              |             |               |
| A: 60/40                                                    | 7.0%           | 7.7%          | 0.90 | -37.6%     | -7.4%     | -3.5%        | 1.6%        | -1.5%         |
| B: 60/40+DAR4020                                            | 9.2%           | 7.9%          | 1.16 | -21.1%     | -4.6%     | -2.6%        | 1.6%        | -0.9%         |
| C: 60/40+Trend                                              | 11.8%          | 9.0%          | 1.31 | -18.3%     | -5.3%     | -3.1%        | 2.1%        | -1.1%         |
| D: 60/40+50/50                                              | 11.4%          | 8.3%          | 1.37 | -15.1%     | -4.4%     | -2.6%        | 1.9%        | -0.9%         |
| C minus B                                                   | 2.6%**         |               |      | 2.7%       | -0.7%     | -0.5%**      | 0.4%**      | -0.2%**       |
| D minus C                                                   | -0.4%**        |               |      | 3.2%       | 1.0%**    | 0.4%**       | -0.2%**     | 0.2%**        |
| D minus A                                                   | 4.4%**         |               |      | 22.5%**    | 3.1%**    | 0.9%**       | 0.2%**      | 0.6%**        |
| <b>Panel B: Portfolio optimizations</b>                     |                |               |      |            |           |              |             |               |
| Minimize risk                                               |                |               |      |            |           |              |             |               |
| E: 60/40+44/00                                              | 8.0%           | 7.4%          | 1.07 | -30.4%     | -6.2%     | -3.1%        | 1.6%        | -1.2%         |
| Maximize Return/Risk                                        |                |               |      |            |           |              |             |               |
| F: 60/40+73/125                                             | 14.7%          | 10.2%         | 1.43 | -1.4%      | -2.8%     | -2.2%        | 2.1%        | -0.6%         |
| Maximize Return/Risk provided overall risk may not increase |                |               |      |            |           |              |             |               |
| G: 60/40+54/36                                              | 9.9%           | 7.7%          | 1.28 | -21.7%     | -5.2%     | -2.8%        | 1.8%        | -1.0%         |

Note: We look at adding protective strategies to 60/40. Protective strategy returns are scaled to 5% volatility based on its ex-post realized volatility over the sample period, including the 50-50 DAR-trend following combination. Panel A looks at simply adding protective strategies to 60/40. Panel B optimizes the weights of the protective strategies looking at Minimum Risk (it then allocates 44% to DAR4020 and nothing to Trend-following) or Maximum Return/Risk portfolios (it then allocates 73% to DAR4020 and 125% to trend-following, and this changes to 54% DAR4020 and 36% trend-following if the total risk should remain the 7.7% risk of the 60/40 portfolio). The Information Ratio (IR) is the ratio for the annualized return and volatility. We look at average cumulative performance over all drawdowns in excess of 20% ('>20% DD') or 2% ('>2% DD'), the percentile of worst 10% returns ('Worst 10%'), and the average performance in months that 60/40 has a positive return ('60/40 up') or a negative return ('60/40 down'), with the latter three presenting average monthly returns. 60/40 returns are funded while the additions are in excess of cash. The sample period is 1800-2021 and \*\* denotes significance at the 1% significance level.

The choice for a 50-50 distribution between DAR4020 and trend-following is arbitrary, as well as the implicit 100% allocation on top of the funded 60/40 portfolio. Perhaps optimal results may be achieved by smaller allocations or deviations from equal weighting, a question especially relevant for asset allocators. Next, we consider explicit portfolio optimization using two main investment objectives for the investor currently invested in 60/40; 1) minimizing overall risk, as measured by volatility, or 2) maximizing return divided by overall risk. We further add a scenario for investment objective 2, where overall risk may not increase. Note that since all our strategies may be implemented through derivatives, leverage is an option. Table 7, panel B contains the results.

When minimizing risk, the 60/40 investor does not invest in trend-following, but adds a 44% allocation to DAR4020 on top of his or her 60/40 portfolio. Overall risk reduces from 7.7% to 7.4%. When maximizing return over risk, a lot of leverage is taken, with 73% allocated to DAR4020, and 125% to trend-following. The resulting portfolio has a return close to 15% per annum, but also runs at considerable risk with a volatility of around 10%. Worst drawdowns are almost gone, but new drawdowns emerge not covered under this metric, since the 60/40 is no longer the main driver of risk in the portfolio, due to highly leveraged exposure to especially trend-following. The final weighting optimization exercise constrains volatility to not exceed that of the 60/40 portfolio and finds an optimal allocation of 54% to DAR4020 and 36% to

trend-following. Irrespective of the investor objectives, we find that DAR4020 is always chosen as a significant investment in the optimal portfolio, improving downside risk measures and overall returns. Further, when seeking more return, trend-following is added.<sup>7</sup>

### 5.3 Other bad states: economic scenarios

Above we have evaluated defensive strategies based on their return properties in negative or tail periods of the 60/40 portfolio. Next, we examine performance conditional upon various economic and market scenarios as they might differ from downside states based on historical asset class returns. Using a deep sample allows us to also look at performance of protective strategies across economic states with sufficient observations. We evaluate inflation, recession, expansion, and combinations.

Table 8 panel A presents performance across recessions and expansions. The 60/40 portfolio earns its returns in expansionary environments, while in recessionary periods it realizes only slightly positive returns on average. Conversely, DAR4020 does especially well in periods characterized by recession, while also returning on average positively in expansion environments. Trend-following does well irrespective of the environment, with expansionary periods strongest. Gold has close to zero returns irrespective of the environment. A combination of both DAR4020 and trend-following has the best performance in recessionary environments, indicating it is a good addition to a 60/40 portfolio which struggles on average in such an environment.

**Table 8: Performance across inflation regimes**

|                                             | Observations | 60/40  | DAR4020 | Trend-following | Gold   | 50-50 DAR4020 –<br>Trend-following |
|---------------------------------------------|--------------|--------|---------|-----------------|--------|------------------------------------|
| <b>Panel A: Expansions and Recessions</b>   |              |        |         |                 |        |                                    |
| Expansions                                  | 1276         | 0.77%  | 0.17%   | 0.54%           | 0.01%  | 0.44%                              |
| Recessions                                  | 488          | 0.04%  | 0.33%   | 0.39%           | −0.03% | 0.45%                              |
| <b>Panel B: Inflation</b>                   |              |        |         |                 |        |                                    |
| <0%                                         | 277          | 0.60%  | 0.23%   | 0.47%           | −0.05% | 0.44%                              |
| 0-2%                                        | 515          | 0.64%  | 0.12%   | 0.50%           | 0.06%  | 0.38%                              |
| 2-4%                                        | 419          | 0.52%  | 0.21%   | 0.40%           | −0.10% | 0.38%                              |
| >4%                                         | 553          | 0.53%  | 0.30%   | 0.58%           | 0.05%  | 0.54%                              |
| <b>Panel C: Inflation during expansions</b> |              |        |         |                 |        |                                    |
| <0%                                         | 116          | 1.26%  | 0.12%   | 0.46%           | −0.02% | 0.36%                              |
| 0-2%                                        | 375          | 0.88%  | 0.07%   | 0.51%           | 0.08%  | 0.36%                              |
| 2-4%                                        | 356          | 0.60%  | 0.18%   | 0.46%           | −0.09% | 0.39%                              |
| >4%                                         | 429          | 0.68%  | 0.26%   | 0.65%           | 0.05%  | 0.57%                              |
| <b>Panel D: Inflation during recessions</b> |              |        |         |                 |        |                                    |
| <0%                                         | 161          | 0.12%  | 0.32%   | 0.48%           | −0.07% | 0.49%                              |
| 0-2%                                        | 140          | 0.02%  | 0.26%   | 0.45%           | 0.01%  | 0.44%                              |
| 2-4%                                        | 63           | 0.01%  | 0.35%   | 0.11%           | −0.19% | 0.28%                              |
| >4%                                         | 124          | −0.02% | 0.41%   | 0.35%           | 0.07%  | 0.47%                              |

Note: Average monthly returns (in nominal terms) of the global portfolio investing 60% in equities and 40% in bonds and the different defensive strategies. Panel A shows performance across NBER recessions and expansions. Panel B considers performance across four regimes of inflation. Panels C and D subset panel A based on NBER expansion and recession indicators. Defensive strategies returns are scaled to 5% volatility based on its ex-post realized volatility over the sample period and in excess of cash, including the equally weighted 50-50 DAR4020 trend-following combination. 60/40 returns are funded. The sample runs from 1875 through 2021, and the observations count reports number of months. The total number of months is 1764, the sum of either Panel A, Panel B, or Panel C and Panel D combined.

<sup>7</sup> We have also performed the portfolio optimization analyses with DAR original (instead of DAR4020) and for different in- and out-of-sample periods. These (unreported) results confirm the above conclusions that DAR and trend-following both have parts to play. Further, in the results presented gold could also be allocated to, but the optimizer never did in line with its poor historical performance. These results are available upon request.

Table 8 Panel B presents performance across inflationary environments, from deflation ('<0%') up to high inflation ('>4%') following the classification of Baltussen et al. (2023). As inflation data is available since 1875, our sample runs from 1875 till 2021. Panel C subsets Panel B into expansionary environments and Panel D subsets Panel B into recessionary environments. Considering Panel B, 60/40 returns are positive across inflationary environments where higher inflation comes at lower returns. DAR4020 and trend-following are both robust to higher inflation, with both strategies realizing the highest returns when 60/40 is relatively low, in high inflation environments. Gold returns are poor across economic environments.

In Panel C, we find the best economic scenario for the 60/40 index, namely economic expansion with low inflation. This is also the environment where DAR4020 performs worst. The higher the inflation, the lower 60/40 returns and the higher DAR4020 returns. Trend-following also does well, with robust performance across inflation and expansion combinations, and best performance when 60/40 does worst, in the high inflation scenario.

Panel D presents different inflationary environments during recessions. Panel D is of great interest as we are considering defensive strategies in one of the toughest scenarios: high inflation with recessions (i.e., stagflation episodes). During recessions the 60/40 index struggles, and the higher the inflation, entering stagflation, the worse the performance, in line with Baltussen et al. (2023). DAR4020 performs well irrespective of the inflation environment during recessions, whereas trend-following does well outside of the moderate inflation '2-4%' subset. The combination of DAR4020 and trend-following comes out as the best performer, with stable positive performance across inflationary states during recessions. Performing when needed most to support the struggling 60/40 portfolio. Overall, both DAR4020 and trend-following prove to be effective performance contributors in the toughest scenarios for the 60/40 portfolio.

## **6. Additional defensive strategies and historical episodes**

So far, we have examined the protection of the 60% equities – 40% bonds portfolio while focusing on the deep sample starting in 1800. Next, we examine protecting a 100% equities reference portfolio and consider shorter samples or subperiods allowing other protective strategies to enter the contest. We evaluate treasuries (section 6.1), the equity factors value and low-risk (section 6.2), the equity factor quality (section 6.3), as well as an equity index put option overlay (section 6.4).<sup>8</sup> In addition, we exploit the depth of our two-century sample to examine how defensive strategies behaved across distinct historical regimes – including pre-Federal Reserve markets, gold-standard eras, major wars, and high-tariff trade-war periods – to test whether defensive properties persist under radically different monetary and institutional systems (section 6.5).

### **6.1 Starting in 1800 and adding Treasuries as defensive strategy**

DAR4020 is constructed by ranking factors on their ex-ante correlation with the benchmark. Now that the benchmark changes from 60% equities 40% bonds to 100% equities it makes sense to define DAR4020 based on the new benchmark, which we label DAR4020-EQ. The results in Table 9 Panel A show that the defensive properties of DAR4020 slightly improve when adjusting DAR4020 to the new benchmark. Of course, in the 60/40 portfolio most

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<sup>8</sup> For 1800-2021, 1800-1984 and 1985-2021 we have compared the results for protecting 60/40 equities/bonds – the benchmark in Cavaglia et al. (2022) – with those for protecting 100% equities – the benchmark in Harvey et al. (2019). Whereas this leads to partly overlapping and partly non-overlapping drawdowns, all the main conclusions are robust to the choice for the benchmark. These results are available upon request.

drawdowns will be driven by the 60% equities, having the larger weight and a higher volatility compared to bonds.

Note that on top of the eight drawdowns in Table 5, with 100% equities we get four new ones that exceed 20%. The two most recent ones are September-November 1987, including Black Monday, and the first quarter of 2020 due to COVID. We further find drawdowns from April 1937 to March 1938, and October 1918 to December 1920. These new drawdowns are all relatively short, and they illustrate the earlier point made about trend-following strategies – i.e., that shorter and large drawdowns can be problematic. The performance of trend-following is negative in the 1914 and ‘Black Monday’ drawdowns. In Table 9, Panel A we see that the average performance of trend-following in the now twelve drawdowns is 12.6%, down from 19.3% for the eight drawdowns in Table 5.

**Table 9: Protecting the 100% equity portfolio**

|                                                                                       | Return p.a. | Stdev p.a. | IR    | >20% DD | >2% DD | Worst 10% | EQ up  | EQ down |
|---------------------------------------------------------------------------------------|-------------|------------|-------|---------|--------|-----------|--------|---------|
| <b>Panel A: 1800-2021</b>                                                             |             |            |       |         |        |           |        |         |
| DAR4020                                                                               | 2.3%        | 5.0%       | 0.46  | 13.1%   | 2.8%   | 1.0%      | -0.04% | 0.60%   |
| DAR4020-EQ                                                                            | 2.5%        | 5.0%       | 0.51  | 13.5%   | 3.0%   | 1.0%      | -0.03% | 0.64%   |
| Trend-following                                                                       | 4.8%        | 5.0%       | 0.97  | 12.6%   | 1.8%   | 0.5%      | 0.42%  | 0.38%   |
| Gold                                                                                  | -0.7%       | 5.0%       | -0.14 | -1.4%   | -1.1%  | -0.4%     | 0.07%  | -0.28%  |
| Treasuries                                                                            | 1.1%        | 5.0%       | 0.22  | -3.5%   | -1.4%  | -0.4%     | 0.28%  | -0.24%  |
| <b>Panel B: 1866-2021– Adding low-risk and value equities as defensive strategies</b> |             |            |       |         |        |           |        |         |
| DAR4020-EQ                                                                            | 2.6%        | 5.0%       | 0.52  | 14.9%   | 3.3%   | 1.2%      | -0.06% | 0.69%   |
| Trend-following                                                                       | 5.7%        | 5.0%       | 1.14  | 12.7%   | 1.9%   | 0.5%      | 0.51%  | 0.41%   |
| Gold                                                                                  | -0.2%       | 5.0%       | -0.04 | 1.3%    | -0.6%  | -0.2%     | 0.11%  | -0.23%  |
| Treasuries                                                                            | 1.0%        | 5.0%       | 0.21  | -1.5%   | -0.9%  | -0.2%     | 0.24%  | -0.17%  |
| Low-risk                                                                              | 0.0%        | 5.0%       | -0.01 | 8.8%    | 2.9%   | 1.1%      | -0.38% | 0.64%   |
| Value                                                                                 | 0.5%        | 5.0%       | 0.11  | 6.4%    | 2.7%   | 0.9%      | -0.28% | 0.60%   |
| <b>Panel C: 1940/7-2021 – Adding quality equities as defensive strategy</b>           |             |            |       |         |        |           |        |         |
| DAR4020-EQ                                                                            | 2.7%        | 5.0%       | 0.55  | 19.3%   | 3.5%   | 1.5%      | -0.10% | 0.86%   |
| Trend-following                                                                       | 5.6%        | 5.0%       | 1.12  | 12.1%   | 1.3%   | 0.4%      | 0.54%  | 0.33%   |
| Gold                                                                                  | 0.6%        | 5.0%       | 0.13  | 2.9%    | -0.4%  | 0.0%      | 0.17%  | -0.17%  |
| Treasuries                                                                            | 1.4%        | 5.0%       | 0.29  | 1.8%    | -0.3%  | -0.2%     | 0.20%  | -0.03%  |
| Low-risk                                                                              | 0.3%        | 5.0%       | 0.06  | 10.4%   | 2.3%   | 1.4%      | -0.32% | 0.68%   |
| Value                                                                                 | 0.6%        | 5.0%       | 0.11  | 6.6%    | 2.2%   | 1.1%      | -0.25% | 0.62%   |
| Quality                                                                               | 3.1%        | 5.0%       | 0.63  | 10.2%   | 2.4%   | 1.1%      | 0.08%  | 0.61%   |
| <b>Panel D: 1986/7-2021 – Adding equity put options as defensive strategy</b>         |             |            |       |         |        |           |        |         |
| DAR4020-EQ                                                                            | 1.6%        | 5.0%       | 0.31  | 17.1%   | 2.6%   | 1.5%      | -0.25% | 0.79%   |
| Trend-following                                                                       | 3.5%        | 5.0%       | 0.69  | 5.4%    | 0.7%   | 0.2%      | 0.37%  | 0.14%   |
| Gold                                                                                  | 0.8%        | 5.0%       | 0.15  | 2.8%    | 0.0%   | 0.2%      | 0.15%  | -0.09%  |
| Treasuries                                                                            | 3.4%        | 5.0%       | 0.69  | 7.6%    | 0.8%   | 0.6%      | 0.25%  | 0.35%   |
| Low-risk                                                                              | 0.2%        | 5.0%       | 0.05  | 12.3%   | 1.9%   | 1.5%      | -0.38% | 0.72%   |
| Value                                                                                 | -0.4%       | 5.0%       | -0.08 | 6.1%    | 1.3%   | 1.1%      | -0.35% | 0.52%   |
| Quality                                                                               | 2.8%        | 5.0%       | 0.56  | 9.8%    | 2.1%   | 1.4%      | -0.02% | 0.67%   |
| Equity put option                                                                     | -2.5%       | 5.0%       | -0.50 | 13.7%   | 1.1%   | 2.0%      | -0.54% | 0.37%   |

Note: We examine protecting a 100% equity reference portfolio. DAR4020 is based on ex-ante correlations with 60/40 equities/bonds, and DAR4020-EQ is based on correlations with 100% equities. We consider annualized return and volatility and their ratio (IR), average cumulative performance over all equity drawdowns in excess of 20% (>20% DD) or 2% (>2% DD), the percentile of worst 10% returns (‘Worst 10%’), and the average performance in months that equities have a positive return (‘EQ up’) or a negative return (‘EQ down’), with the latter three presenting average monthly returns. Each protective strategy is scaled to 5% volatility based on its ex-post realized volatility over the sample period to make them comparable to each other. Returns are in excess of cash. For the sample period 1800-2021 there are 12 drawdowns (‘DD’) exceeding 20%. This reduces to 10 drawdowns when starting in 1866, 6 when starting in 1940/7, and 4 when starting in 1986/7.

Given the new benchmark it makes sense to include global developed market treasuries (USD hedged) as a protective strategy. This made less sense when treasuries themselves were part of

the benchmark. The average performance of Treasuries during the 12 largest equity drawdowns from 1800 is a negative  $-3.5\%$  over cash. In fact, Treasuries make most of their returns in positive equity months. This is perhaps surprising given that many consider Treasuries as a safe haven in crisis periods. In Panel D of Table 9, discussed later, we do indeed see that in the 4 large equity drawdowns from 1986/7 onwards Treasuries did on average have a positive performance.

In line with Harvey et. al (2019) we conclude that treasuries are a subpar hedge as they are historically unreliable, even though we consider global bonds and Harvey et. al used only U.S. treasuries for their analysis. We further find that the defensive characteristics of the DAR4020 strategy is robust to a reference portfolio of 60/40 equity/bond as well as a full equity portfolio.

## **6.2 Starting in 1866 and adding low-risk and value as defensive strategies**

Next, we move to 1866, the point where equity selection strategies become available that do not need accounting information: low-risk, based on past 36-months return volatilities, and value based on dividend yield obtained from Baltussen, van Vliet and Van Vliet (2023). Both factors are constructed dollar-neutral, hence we do not lever the low or high volatility portfolios to beta-neutrality. Table 9, Panel B shows that both have modest average returns at  $0.0\%$  and  $0.5\%$ , respectively. It does highlight that over this sample, low-risk stocks have achieved about the same total return as high-risk stocks, a remarkable result, and in line with evidence of a low-risk effect over more recent samples (see Blitz, Van Vliet, and Baltussen, 2020). Both low-risk and value have good defensive properties over our deep sample, making strong positive returns in negative equity months, on average returning  $8.8\%$  and  $6.4\%$ , respectively, in the 10 large equity drawdowns since 1866. However, when competing with DAR4020-EQ, DAR scores better on all dimensions with a higher total return and better defensive properties. Versus trend-following performance characteristics are more nuanced, where trend-following does better in the worst drawdowns, but these strategies do better in modest drawdowns and generally in equity down states.

## **6.3 Starting in 1940/7 and adding quality as defensive strategy**

We further move the starting point forward to 1940/7, when quality based on profitability becomes available. Results are presented in Table 9, Panel C. Quality has attractive returns and strong defensive properties, in line with the flight-to-quality paradigm, providing a positive out-of-sample test for the positive recommendation for quality by Harvey et al. (2019). Quality is as strong as low-risk and value in negative equity months, and in terms of average performance over the remaining six large equity drawdowns it is not far behind trend-following and low-risk. However still for this shorter sample, DAR4020-EQ has clearly the strongest defensive properties. We mentioned earlier that DAR4020 benefits from an expanding universe, especially when factors are added with either a strong positive or a strong negative correlation with the benchmark. Here DAR4020 is benefitting from the addition of quality to the opportunity set as it is taken along in the correlation ranking within DAR4020. For this shorter sample quality realizes similar defensive properties to low-risk, but at a healthy positive total return. It outperforms some defensive characteristics of trend-following, but not in the worst drawdowns and at a much-reduced return profile.

## **6.4 Starting in 1986/7 and adding equity put options as defensive strategy**

Finally, we move the starting point to 1986/7, from where we can assess the attractiveness of equity index put options. We make use of the CBOE S&P 500 5% put protection index.<sup>9</sup> It

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<sup>9</sup> Bloomberg ticker: PPUT Index.

combines S&P 500 equity returns with buying a monthly 5% out of the money put option as a hedge. By subtracting the S&P 500 index returns,<sup>10</sup> we are left with the excess returns of the put option overlay.<sup>11</sup> We constrain ourselves to this specific option type as data is readily available and it gives a good baseline for drawdown protection at capping monthly losses at 5%. At-the-money puts deliver only limited convexity and therefore are not included, as in Ilmanen et al. (2021). The results are presented in Table 9, Panel D. The put option provides positive returns during the most extreme down states but runs at a negative annual return of −2.5% (when scaled to 5% realized volatility), over this relatively short sample. This aligns with earlier results of Ilmanen et al. (2021) that put options provide protective payoffs but come at the cost of a negative carry.

In severe drawdowns (>20%), the strongest performer is DAR4020 by a wide margin, followed by low-risk and the put option. The put option performance is perhaps less strong than one might expect, since it requires drawdowns to be steep within a month. In a scenario where the market sells off 4% per month, the market loss can build up to be massive while the put option pays out zero, since the monthly loss of 5% is never achieved before rolling into a new contract. In drawdowns deeper than 2%, the put option drops from the top performers to number 5 in our set, while DAR4020, quality, and low-risk perform best over this more limited subsample. Considering the averages across all equity down months, we find best performance for DAR4020 around 0.80% followed by low-risk and quality at around 0.70%, while put option strategies perform materially worse. Albeit more limited than most other defensive strategies, trend-following also provides defensive properties over this subsample, in line with earlier results by Ilmanen et al. (2021), who find that trend-following provides protective payoffs similar to put options but with a positive carry.

We conclude that even for this recent window which includes a large bond bull market, treasuries were not the best defensive asset to have. DAR4020 is the top defensive performer, and the put option, although effective in steep drawdowns, loses the race when considering the wider set of drawdowns in combination with its negative average return per annum. On average, running a structural put option overlay quickly becomes prohibitively costly, especially compared to DAR4020 or trend-following.

## 6.5 Historical regimes

Before this point, we have evaluated defensive strategies across various subsamples and economic scenarios, with sample choices driven by data availability or the identification of in-sample and out-of-sample periods. We now turn to historical regimes to examine whether defensive properties persist under vastly different institutional and monetary settings.

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<sup>10</sup> Bloomberg ticker: SPXT Index as of 1988/2. For 1986/7 up to 1988/2 we use the SPX Index and add a proxy for monthly dividends as the average monthly dividend yield over the first 12 months that we have the SPXT Index available, 1988/2 through 1989/1.

<sup>11</sup> A shortcoming is that the equity index we consider includes developed markets outside of the U.S., while the option is based on U.S. equities. Considering the large weight of U.S. equities within the global equity index and high correlations across global equities, this option is deemed a representative candidate for our purpose.

**Table 10: Protecting the 100% equity portfolio – Interesting deep history episodes**

|                                                                                          | Return p.a. | Stdev p.a. | IR    | >20% DD | >2% DD  | Worst 10% | EQ up  | EQ down |
|------------------------------------------------------------------------------------------|-------------|------------|-------|---------|---------|-----------|--------|---------|
| <b>Panel A: 1800-1913 – Pre-Federal Reserve era in U.S.</b>                              |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 2.2%        | 5.0%       | 0.45  | 10.1%   | 2.52%   | 0.57%     | 0.01%  | 0.51%   |
| Trend-following                                                                          | 4.1%        | 5.2%       | 0.79  | 15.0%   | 1.82%   | 0.48%     | 0.34%  | 0.35%   |
| Gold                                                                                     | -2.0%       | 3.0%       | -0.66 | -15.5%  | -1.57%  | -0.72%    | -0.06% | -0.36%  |
| Treasuries                                                                               | 1.0%        | 5.7%       | 0.18  | -15.5%  | -2.59%  | -0.77%    | 0.37%  | -0.41%  |
| <b>Panel B: 1873-1896 – Long Depression</b>                                              |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 1.3%        | 4.8%       | 0.28  |         | 1.26%   | 0.52%     | -0.02% | 0.32%   |
| Trend-following                                                                          | 4.2%        | 4.3%       | 0.96  |         | 1.27%   | 0.07%     | 0.44%  | 0.19%   |
| Gold                                                                                     | -1.8%       | 0.9%       | -2.0  |         | -0.57%  | -0.29%    | -0.10% | -0.23%  |
| Treasuries                                                                               | 1.3%        | 2.0%       | 0.63  |         | -0.63%  | -0.27%    | 0.24%  | -0.12%  |
| Low-risk                                                                                 | 0.6%        | 6.1%       | 0.10  |         | 5.72%   | 1.71%     | -0.43% | 0.85%   |
| Value                                                                                    | 1.5%        | 4.9%       | 0.31  |         | 4.74%   | 1.58%     | -0.27% | 0.78%   |
| <b>Panel C: 1815-1870 – Post-Napoleonic Wars and early gold convertibility era</b>       |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 2.6%        | 5.3%       | 0.49  | 10.1%   | 2.71%   | 0.62%     | 0.05%  | 0.51%   |
| Trend-following                                                                          | 3.5%        | 5.7%       | 0.62  | 15.0%   | 1.99%   | 0.48%     | 0.25%  | 0.36%   |
| Gold                                                                                     | -2.2%       | 4.2%       | -0.52 | -15.5%  | -2.48%  | -1.20%    | 0.01%  | -0.52%  |
| Treasuries                                                                               | 0.6%        | 6.2%       | 0.09  | -15.5%  | -5.06%  | -1.63%    | 0.50%  | -0.78%  |
| <b>Panel D: 1861-1865 – American Civil War financial impacts era</b>                     |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 2.4%        | 6.1%       | 0.40  | 5.1%    | 4.10%   | 0.99%     | -0.05% | 0.85%   |
| Trend-following                                                                          | 6.1%        | 6.0%       | 1.00  | -5.1%   | -1.34%  | -0.90%    | 0.79%  | -0.21%  |
| Gold                                                                                     | 2.3%        | 12.2%      | 0.19  | -27.0%  | -13.52% | -6.73%    | 1.36%  | -2.76%  |
| Treasuries                                                                               | 0.0%        | 4.0%       | 0.00  | -1.5%   | -0.19%  | -0.78%    | 0.12%  | -0.31%  |
| <b>Panel E: 1870-1900 – Gilded Age</b>                                                   |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 0.9%        | 4.5%       | 0.21  |         | 1.28%   | 0.37%     | -0.07% | 0.34%   |
| Trend-following                                                                          | 4.3%        | 4.3%       | 1.00  |         | 1.21%   | 0.11%     | 0.45%  | 0.18%   |
| Gold                                                                                     | -1.9%       | 1.3%       | -1.45 |         | -0.51%  | -0.35%    | -0.10% | -0.26%  |
| Treasuries                                                                               | 0.9%        | 2.3%       | 0.37  |         | -0.89%  | -0.33%    | 0.23%  | -0.21%  |
| Low-risk                                                                                 | -0.7%       | 6.0%       | -0.11 |         | 4.93%   | 1.37%     | -0.56% | 0.82%   |
| Value                                                                                    | 1.0%        | 4.7%       | 0.21  |         | 4.02%   | 1.22%     | -0.31% | 0.78%   |
| <b>Panel F: 1870-1914 – Classical Gold Standard era</b>                                  |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 2.1%        | 4.4%       | 0.48  |         | 2.50%   | 0.57%     | -0.02% | 0.49%   |
| Trend-following                                                                          | 4.8%        | 4.4%       | 1.09  |         | 1.35%   | 0.25%     | 0.48%  | 0.29%   |
| Gold                                                                                     | -1.9%       | 1.1%       | -1.71 |         | -0.70%  | -0.27%    | -0.12% | -0.22%  |
| Treasuries                                                                               | -0.1%       | 2.4%       | -0.06 |         | -1.63%  | -0.46%    | 0.15%  | -0.27%  |
| Low-risk                                                                                 | -0.7%       | 5.6%       | -0.12 |         | 4.80%   | 1.30%     | -0.54% | 0.74%   |
| Value                                                                                    | 1.1%        | 4.4%       | 0.24  |         | 3.96%   | 1.04%     | -0.29% | 0.71%   |
| <b>Panel G: 1914-1918 – World War I</b>                                                  |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 3.8%        | 6.1%       | 0.62  |         | 2.61%   | 0.65%     | 0.07%  | 0.64%   |
| Trend-following                                                                          | 5.5%        | 6.0%       | 0.91  |         | 2.90%   | 1.10%     | 0.27%  | 0.70%   |
| Gold                                                                                     | -1.8%       | 0.1%       | -21.6 |         | -0.70%  | -0.16%    | -0.15% | -0.16%  |
| Treasuries                                                                               | -4.5%       | 4.7%       | -0.97 |         | -2.47%  | -0.18%    | -0.22% | -0.58%  |
| Low-risk                                                                                 | -3.6%       | 4.7%       | -0.77 |         | 1.50%   | 0.18%     | -0.70% | 0.22%   |
| Value                                                                                    | -2.1%       | 5.1%       | -0.42 |         | 2.41%   | -0.01%    | -0.53% | 0.29%   |
| <b>Panel H: 1919-1939 – Interwar period and Great Depression</b>                         |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 3.5%        | 5.9%       | 0.58  | 10.7%   | 5.27%   | 1.40%     | 0.01%  | 0.67%   |
| Trend-following                                                                          | 6.9%        | 4.6%       | 1.51  | 17.1%   | 7.27%   | 1.40%     | 0.51%  | 0.67%   |
| Gold                                                                                     | 0.2%        | 4.2%       | 0.05  | -1.8%   | -1.24%  | -0.11%    | 0.09%  | -0.09%  |
| Treasuries                                                                               | 2.3%        | 4.0%       | 0.56  | 2.7%    | -1.48%  | -0.52%    | 0.39%  | -0.09%  |
| Low-risk                                                                                 | 0.3%        | 6.0%       | 0.05  | 16.7%   | 5.81%   | 1.29%     | -0.48% | 0.72%   |
| Value                                                                                    | -0.3%       | 6.1%       | -0.05 | 5.4%    | 4.11%   | 1.05%     | -0.42% | 0.51%   |
| <b>Panel I: 1816-1832, 1843-1846, 1861-1913, 1930-1945 – High tariffs and trade wars</b> |             |            |       |         |         |           |        |         |
| DAR4020-EQ                                                                               | 2.0%        | 4.9%       | 0.40  | 4.3%    | 2.96%   | 0.84%     | -0.01% | 0.48%   |
| Trend-following                                                                          | 4.6%        | 4.6%       | 0.99  | -1.2%   | 1.66%   | 0.28%     | 0.42%  | 0.32%   |
| Gold                                                                                     | -1.3%       | 4.9%       | -0.35 | -13.5%  | -1.69%  | -0.70%    | 0.03%  | -0.37%  |
| Treasuries                                                                               | 1.1%        | 4.0%       | 0.26  | 1.0%    | -1.65%  | -0.42%    | 0.31%  | -0.31%  |
| Low-risk                                                                                 | -0.8%       | 5.8%       | -0.13 | 12.1%   | 3.52%   | 1.45%     | -0.55% | 0.73%   |
| Value                                                                                    | 0.2%        | 4.9%       | 0.05  | 5.1%    | 2.55%   | 0.97%     | -0.34% | 0.60%   |

Note: See Table 9, with the exception that we do not rescale the returns of each protective strategy to 5% volatility for each subperiod. A >20% Drawdown period is only included if it falls completely in the stated period. In Panel I results for Low-Risk and Value are based on the subsample where they are available, i.e. from 1866.

Specifically, we consider: (i) the pre-FED period (1800-1913) to assess defensive strategies in an environment without modern central bank interventions; (ii) the Long Depression (1873-1896) of deflation and slow growth; (iii) the post-Napoleonic wars and early gold convertibility era (1815-1870) marked by transitioning monetary systems, early industrialization, and commodity-backed currencies; (iv) the American Civil War (1861-1865), characterized by wartime economic distortions and currency instability; (v) the Gilded Age (1870-1900) a growth-oriented but volatile era with limited financial regulation; (vi) the Classical Gold Standard era (1870-1914) when gold itself served as the monetary base; (vii) World War I (1914-1918) testing safe-haven assets during global conflict and mobilization; and (viii) the Interwar period and Great Depression (1919-1939), dominated by monetary instability, banking failures, the deepest economic downturn, and major policy reforms beginning with the New Deal in 1933. Finally, we examine periods of high tariffs or trade wars evident more recently in 2025 but largely absent the decades before. We define historical periods of high trade tariffs based on the U.S. effective tariff rate (customs duty revenue divided by goods imports) from Yale BudgetLab.<sup>12</sup> These periods include 1816-1832, 1843-1846, 1861-1913, and 1930-1945.<sup>13</sup>

Table 10 summarizes defensive strategy performances for each of the above historical episodes. Across regimes, DAR4020 consistently delivered positive returns and downside protection, even in eras without modern monetary policy (pre-Fed) or fiat currencies (gold-standard environments). Trend-following is the strongest strategy in terms of overall returns with the highest information ratios across regimes. It provided strong defensive characteristics except during the Civil War, which also weakened results in the overlapping high-tariff and trade-war periods and showed muted downside performance during the Long Depression. Overall, its defensive performance is less consistent than DAR4040 across historical regimes. Gold is weak again across samples, both from an overall return perspective and a defensive perspective. This includes the Classical Gold Standard era. Treasury returns were occasionally positive – such as during the Interwar and Great Depression eras – but generally failed to hedge equity losses, producing negative results on defensive metrics. For the periods where equity factors are available (from 1866 onward), low-risk and value both emerge as relatively reliable defensive strategies. Value tends to outperform in overall returns, while low-risk exhibits stronger downside protection. Each shows resilience in defensive performances during historical regimes, although both experienced weak absolute returns in certain crises (for instance, low-risk -3.6% p.a. and value -2.1% p.a. during World War I).

These patterns indicate that DAR defensive payoffs especially have been regime-agnostic, working across every regime. For practitioners, this analysis offers a two-century

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<sup>12</sup> <https://budgetlab.yale.edu/>.

<sup>13</sup> Below we highlight further details on these historical high-tariff periods:

1816 – 1832: The Early Republic protectionist era, when effective tariffs averaged above 30%, aimed at raising revenue to finance war debts and shielding emerging manufacturing industries.

1843 – 1846: A tariff spike episode reflecting renewed protection for domestic manufacturing following fiscal shortfalls and competition from Britain. Effective tariffs on dutiable imports rose to roughly 35–40% before being sharply reduced by the liberalizing 1846 Walker Tariff.

1861 – 1913: In the Civil War the U.S. implemented the Morrill Tariff in 1861, raising average tariff rates on dutiable commodities to approximately 47% to generate revenue for the Civil War. From the post-Civil War era (which was 1861–1865) to World War I, tariffs remained high to protect flourishing industries. Tariffs on dutiable imports averaged between 30% and 50%.

1930 – 1945: Following the 1929 stock market crash, countries sought to protect their domestic industries and jobs. In 1930 the Smoot-Hawley Tariff Act raised U.S. tariffs on over 20,000 imported goods to historically high levels. Average tariffs on dutiable imports were pushed to roughly 45%. Instead of helping, it triggered a trade war as other countries like Canada, the UK, France, and Germany retaliated with their own tariffs and trade barriers.

historical stress-test showing that defensive approaches remain effective across a wide variety of historical economic regimes.

## **7. Conclusion**

Balanced global equity/global bond portfolios are vulnerable to severe drawdowns. We examine a wide spectrum of downside-protection, or defensive, strategies over more than 220 years of data – covering many years where traditional equity-bond portfolios suffer, across a range of economic conditions, institutional systems, and distinct historical regimes.

Our results show that traditional defensive equity factors like low-risk, quality but also value offer meaningful downside protection, whereas gold and put options prove less drawdown-, performance-, or cost-effective. The Defensive Absolute Return (DAR) approach and trend-following strategies emerge as the most consistently robust defensive mechanisms across two centuries of evidence. DAR achieves its protection by going long factors with negative correlations to a 60/40 portfolio and short those with positive correlations, thereby generating a negative beta to the traditional balanced portfolio at theoretically zero cost.

We further propose an enhanced implementation, DAR4020, which adds a return component from factor premia while preserving its defensive nature. DAR4020 complements trend-following by responding immediately at the onset of market declines and performing particularly well in large, short-lived drawdowns. A combination of DAR4020 and trend-following provides a better defensive portfolio than either strategy in isolation, across time periods and economic states. Overall, we believe that investors can improve the defensive properties of traditional portfolios by considering the historical insights and long-run evidence on defensive strategies presented in this study.

## **References**

- Accominotti, Olivier, and David Chambers. 2016. If you're so smart: John Maynard Keynes and currency speculation in the interwar years. *Journal of Economic History* 76 (2), 342-386.
- Asness, Cliff, Tobias Moskowitz, and Lasse H. Pedersen. 2013. Value and Momentum Everywhere. *Journal of Finance* 68 (3), 929-985.
- Babu, Abhilash, and Brendan Hoffman, Ari Levine, Yao Hua Ooi, Sarah Schroeder, and Erik Stamelos. 2020. You Can't Always Trend When You Want. *The Journal of Portfolio Management* 46 (4), 1-17.
- Baltussen, Guido, Laurens Swinkels, and Pim van Vliet. 2021. Global Factor Premiums. *Journal of Financial Economics* 142 (3), 1128-1154.
- Baltussen, Guido, Laurens Swinkels, Bart van Vliet, and Pim van Vliet. 2023. Investing in Deflation, Inflation, and Stagflation Regimes. *Financial Analysts Journal* 79 (3), 5-32.
- Baltussen, Guido, Bart van Vliet, and Pim van Vliet. 2023. The Cross-Section of Stock Returns before CRSP. SSRN working paper.
- Blitz, David, Pim van Vliet, and Guido Baltussen, 2020. The Volatility Effect Revisited. *The Journal of Portfolio Management* 46 (2), 45-63.
- Brown Jr, William O., J. Harold Mulherin, and Marc D. Weidenmier. 2008. Competing with the New York stock exchange. *The Quarterly Journal of Economics*, 123 (4), 1679-1719.
- Cavaglia, Stefano, John Hua Fan, and Zhenping Wang. 2022. Portable Beta and Total Portfolio Management. *Financial Analysts Journal* 78 (3), 49-69.
- Chambers, David, and Elroy Dimson. 2016. Financial market history: reflections on the past for investors today. CFA Research Institute Foundation.
- Erb, Claude B., and Campbell R. Harvey. 2013. The Golden Dilemma. *Financial Analysts Journal* 69 (4), 10-42.
- Fama, Eugene F., and Kenneth R. French. 1992. The Cross-Section of Expected Stock Returns. *Journal of Finance* 47 (2), 427-465.
- Fama, Eugene F., and Kenneth R. French. 1993. Common Risk Factors in the Returns on Stocks and bonds. *Journal of Financial Economics* 33 (1), 3-56.
- Frazzini, Andrea, and Lasse H. Pedersen. 2014. Betting Against Beta. *Journal of Financial Economics* 111 (1), 1-25.
- Harrison, Paul. 1998. Similarities in the distribution of stock market price changes between the eighteenth and twentieth centuries. *Journal of Business* 71, 55-79.
- Harvey, Campbell R., Edward Hoyle, Sandy Rattray, Matthew Sargaison, Dan Taylor, and Otto Van Hemert. 2019. The Best of Strategies for the Worst of Times: Can Portfolios Be Crisis Proofed? *Journal of Portfolio Management* 45 (5), 1-22.
- Jones, Charles. M. 2002. A century of stock market liquidity and trading costs. *Working paper*, available at SSRN 313681.
- Koudijs, Peter. 2016. The boats that did not sail: asset price volatility in a natural experiment. *Journal of Finance* 71 (3), 1185-1226.
- Neal, Larry. 1985. Integration of international capital markets: quantitative evidence from the eighteenth to twentieth centuries. *Journal of Economic History* 45 (2), 219-226.

- Neal, Larry. 1987. The integration and efficiency of the London and Amsterdam stock markets in the eighteenth century. *Journal of Economic History* 47(1), 97-115.
- Poitras, Geoffrey. 2009. From Antwerp to Chicago: the history of exchange traded derivative security contracts. *Revue d'Histoire des Sciences Humaines* 20(1), 11-50.
- Van Hemert, Otto, Mark Ganz, Campbell R. Harvey, Sandy Rattray, Eva Sanchez Martin, and Darrel Yawitch, 2020. Drawdowns. *Journal of Portfolio Management* 46 (8), 34–50.
- Hurst, B., Y. H. Ooi, and L. H. Pedersen. 2017. A Century of Evidence on Trend-Following Investing. *The Journal of Portfolio Management* 44 (1): 15–29.
- Goulding, Christian. L., Campbell R. Harvey, and Michele G. Mazzoleni, 2023a. Momentum Turning Points. *Journal of Financial Economics* 149 (3): 378–406.
- Goulding, Christian. L., Campbell R. Harvey, and Michele G. Mazzoleni, 2023b. Breaking Bad Trend. *Financial Analysts Journal* 80 (1), 84-98.
- Ilmanen, Antti, Ashwin Thaper, Harsha Tummala, and Dan Villalón. 2021. Tail Risk Hedging: Contrasting Put and Trend Strategies. *Journal of Systematic Investing* 1 (1), 111-124.
- Jegadeesh, Narasimhan, and Sheridan Titman. 1993. Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *Journal of Finance*, 48 (1), 65-91.
- Jones, Charles M. 2002. A century of stock market liquidity and trading costs. *Working paper*; available at SSRN 313681.
- Koijen, Ralph S.J., Tobias J. Moskowitz, Lasse Heje Pedersen, Evert B. Vrugt. 2018. Carry. *Journal of Financial Economics* 127 (2), 197-225.
- Moskowitz, Tobias J., Yao Hua Ooi, and Lasse Heje Pedersen. 2012. Time Series Momentum. *Journal of Financial Economics* 104 (2), 416-441.
- Novy-Marx, Robert. 2013. The Other Side of Value: The Gross Profitability Premium. *Journal of Financial Economics* 108 (1), 1-28.
- Wahal, Sunil. 2019. The Profitability and Investment Premium: Pre-1963 evidence. *Journal of Financial Economics* 131 (2), 362-377.

## **Appendix A: Dataset construction**

In this Appendix we outline the data sources and factor definitions used in this study, mostly building upon Baltussen, Swinkels and Van Vliet (2021), Baltussen, Swinkels, Van Vliet and Van Vliet (2023, henceforth BSV), and Baltussen, Van Vliet, and Van Vliet (2023). For further information we refer to the studies.

**Asset class returns.** Equity and bond market returns are sourced from Datastream or Bloomberg, spliced with Global Financial Data (GFD), as in Baltussen, Swinkels and Van Vliet (2021). We use global equity market returns expressed in USD from MSCI World, and before existence use global value-weighted equity market returns from GFD (ticker TRWLDM) and in case not available, weighted market returns across key developed markets. For the global government bond returns we use returns on the Bloomberg Barclays Global Treasury index, which we splice before inception with GDP-weighted bond returns across U.S., U.K., German, France, and Japanese bond markets. Bond return data is from Bloomberg and GFD, while GDP data is from the Maddison Project Database (Maddison, 2006). GDP data is available from April 1871, global bonds are equally weighted before across the markets outlined above based on the availability shown in the Online Appendix of BSV. All bond returns are hedged to USD, as common in practice as currency risk is a very substantial driver of relative risk in bond portfolios. Cash returns are returns on short-dated U.S. Treasury Bills obtained from Kenneth French website and before its inception from Jeremy Siegel. The 60/40 portfolio is constructed as 60% market cap weighted equities and 40% global bonds (hedged to USD). Gold prices and returns are from GFD and Bloomberg.

**Equity Factor returns.** For the pre-1927 data, we use data from Baltussen, Van Vliet and Van Vliet (2023), for the post-1926 data, we use the Kenneth French on-line data library. Baltussen, Van Vliet and Van Vliet (2022) have compiled their data from several sources in order to obtain a reliable and historically extensive dataset. The sample covers 61 years of data on monthly stock prices, dividend yields and market capitalizations for all major stocks traded on the NYSE, NY Curb, and regional exchanges. The sample spans the period from January 1866 through December 1926 and is at the monthly frequency. They build their dataset from the Commercial and Financial Chronicle (CFC, which was also used to build the CRSP sample as of 1926) and Global Financial Data (GFD). Further, we splice Quality in U.S. equities using the data from Wahal (2019), who extends history to 1940 for investment and profitability, as available on Wahal's website.

The equity factors are constructed as follows:

- Value:
  - 1875-1926: Dividend yield. Source: Baltussen, Van Vliet, and Van Vliet (2022).
  - 1927-2021: Book-to-market (HML). Source. Kenneth French on-line data library.
- Momentum:
  - 1875-1926: Past 12-1 months total return. Source: Baltussen, Van Vliet, and Van Vliet (2022)
  - 1927-2021: Past 12-1 months total return (MOM). Source. Kenneth French on-line data library.
- Low risk:
  - 1875-1926: Past 36-months beta. Beta neutral long-short portfolio. Source: Baltussen, Van Vliet, and Van Vliet (2022).
  - 1927-2021: Past 36-months volatility. Volatility neutral long-short portfolio. Source. [www.paradoxinvestor.com](http://www.paradoxinvestor.com)
- Quality (50% Profitability & 50% Investment):

- 1875-1939: Not available, because companies did not have (standardized) accounting data, see Wahal (2019) and Baltussen, Van Vliet, and Van Vliet (2023).
- 1940-1962: Operating profitability, defined as revenues minus cost of goods sold, minus selling, general, and administrative expenses, minus interest expense, scaled by book equity. Investments, defined as the change in total assets from the fiscal year ending in year t-2 to the fiscal year ending in t-1, divided by t-2 total assets. Source: Wahal (2019).
- 1963-2021: Operating profitability (RMW) and Investments (CMA), defined the same as above. Source: Kenneth French on-line data library.

**Global Factor Premiums.** For global ('cross-asset') factor premium we use the dataset and definitions from BSV. They source price and return data of equity futures and indices from Bloomberg, Datastream, and GFD. Their primary source is the futures from Bloomberg, with gaps filled in by Datastream data and spliced before futures inception with index-level data. They backfill these data with equity index-level data downloaded from GFD and obtain dividend yields from the same sources. For Carry, they use the spot, front futures, and second futures prices. Before they have data on futures, they reconstruct the monthly implied carry as if these markets had listed futures using the regression methodology on the difference between total return and price indices. The markets considered are spread around the globe and cover the major developed markets with substantial data history.

Bond futures price and return data are sourced from Bloomberg and spliced with bond index-level data from Datastream, backfilled before inception with GFD. From the same sources, they obtain yields and inflation data, the latter extended when possible with data from Macrohistory.net. The markets considered are the major developed bond markets around the globe.

Currency forward and spot prices are primarily from Datastream, spliced with Bloomberg data and GFD. Purchasing power parity data are obtained from the OECD website and, before 1971, with data from Macrohistory.net. The main measure for the financing rates is short-term London Inter-bank Offered Rate (LIBOR) rates (sourced from Bloomberg and Datastream), spliced with (in order of usage) Eurodollar rates from Datastream, short-term Treasury bill rates and commercial paper yields from GFD, and short rates from Macrohistory.net and, for the U.S., with data from Jeremy Siegel. When all are unavailable, they splice with lagged Treasury bill returns.

Commodity futures price and return data are sourced from Bloomberg, spliced with monthly commodity futures data from CBOT annual reports (1877–1962) obtained from TwoCenturies.com. For the commodity value measure, they use commodity spot prices from Bloomberg and Datastream, spliced with spot data from GFD and TwoCenturies.com. For Carry, they use the front futures and second futures prices. The main contracts based on their general usage and liquidity are included.

The global factor premiums are constructed as follows:

- Value: For equities, the dividend-to-price ratio (D/P) defined as past 12-month dividend payment by the current price, for government bonds the Real yield (bond yield minus inflation), for currencies an equally weighted combination of absolute and relative purchasing power parity, and for commodities, the 5-year reversal in spot prices.
- Momentum/Trend: Time-series Momentum that takes a long (short) position if the 12-1 months return is positive (negative).
- Low-risk: Past 36-month beta with respect to asset class. Beta-neutral long-short portfolio.
- Carry: For equities the excess implied dividend yield priced into the futures versus spot contract, for government bonds the slope of the yield curve defined as the 10-year yield

minus the short rate, for currencies the short-term yield differential, and for commodities the slope of the futures curve.

Assets included in the global factor premiums strategies are included once available in the databases mentioned above and surpassing the various data quality filters and checks utilized by BSV. Table A.1 below summarizes per main asset class each asset / index included in the global factor premiums including their start dates. For example, equity market returns on the U.S. and U.K. markets are included as of 1800, followed by France as of 1803, Australia as of 1835, Germany as of 1837, and others after. Bond market returns on the U.S., U.K. and France are included as of 1800, the Netherlands as of 1814, other European markets after, Australia as of 1859, and Japan as of 1880. Currencies (all relative to the USD) start in 1800 for the GBP, 1801 for the FRF, followed by other European markets after, the AUD as of 1825, and the JPY as of 1880. Commodity series start as of 1877, and hence global commodity factor premiums as of 1878. We refer to Online Appendix A of BSV for further details, including data series tickers and summary statistics, of the global factor premiums and their underlying markets.

**Table A.1: Starting years of series included in the global factor premiums universe**

| <b>Equity markets</b> | <b>Index</b> | <b>Return</b> | <b>Value</b> | <b>Carry</b> |
|-----------------------|--------------|---------------|--------------|--------------|
| U.S.                  | S&P500       | 1800          | 1800         | 1800         |
| U.K.                  | FTSE100      | 1800          | 1800         | 1800         |
| France                | CAC          | 1802          | 1802         | 1806         |
| Australia             | ASX200       | 1834          | 1834         | 1834         |
| Germany               | DAX          | 1835          | 1835         | 1840         |
| Spain                 | IBEX         | 1881          | 1881         | 1881         |
| Japan                 | TOPIX        | 1886          | 1886         | 1890         |
| Netherlands           | AEX          | 1892          | 1892         | 1896         |
| Canada                | TSE60        | 1900          | 1900         | 1900         |
| Sweden                | OMX          | 1902          | 1902         | 1906         |
| Italy                 | MIB          | 1925          | 1925         | 1929         |
| South Korea           | KOSPI200     | 1965          | 1965         | 1969         |
| Switzerland           | SMI          | 1966          | 1966         | 1971         |
| Hong Kong SAR         | HANGSENG     | 1968          | 1968         | 1969         |

| <b>Bond markets</b> | <b>Index</b> | <b>Return</b> | <b>Value</b> | <b>Carry</b> |
|---------------------|--------------|---------------|--------------|--------------|
| U.S.                | 10Y          | 1800          | 1872         | 1800         |
| U.K.                | 10Y          | 1800          | 1872         | 1800         |
| France              | 10Y          | 1800          | 1872         | 1800         |
| Netherlands         | 10Y          | 1814          | 1872         | 1814         |
| Norway              | 10Y          | 1822          | 1872         | 1822         |
| Germany             | 10Y          | 1832          | 1869         | 1832         |
| Belgium             | 10Y          | 1848          | 1872         | 1848         |
| Sweden              | 10Y          | 1853          | 1872         | 1854         |
| Australia           | 10Y          | 1857          | 1872         | 1857         |
| Italy               | 10Y          | 1862          | 1872         | 1862         |
| Denmark             | 10Y          | 1864          | 1872         | 1864         |
| Japan               | 10Y          | 1880          | 1880         | 1880         |
| Spain               | 10Y          | 1881          | 1881         | 1881         |
| Canada              | 10Y          | 1900          | 1900         | 1900         |
| Switzerland         | 10Y          | 1900          | 1900         | 1900         |
| New Zealand         | 10Y          | 1923          | 1923         | 1923         |

| <b>Commodities</b>  | <b>Return</b> | <b>Value</b> | <b>Carry</b> | <b>Commodities</b> | <b>Return</b> | <b>Value</b> | <b>Carry</b> |
|---------------------|---------------|--------------|--------------|--------------------|---------------|--------------|--------------|
| Wheat               | 1877          | 1877         | 1877         | Cocoa              | 1967          | 1967         | 1967         |
| Oats*               | 1877          | 1877         | 1885         | Kansas Wheat       | 1970          | 1983         | 1970         |
| Corn                | 1877          | 1877         | 1885         | Coffee             | 1973          | 1973         | 1973         |
| Lard*               | 1877          | 1889         | 1877         | Gold               | 1975          | 1975         | 1975         |
| Pork*               | 1877          | 1975         | 1877         | Silver             | 1975          | 1975         | 1975         |
| Short Ribs*         | 1885          | 1889         | 1885         | Oil (WTI)          | 1983          | 1983         | 1983         |
| Rye*                | 1885          | 1885         | 1886         | Heating Oil        | 1986          | 1986         | 1986         |
| Barley*             | 1885          | 1885         | 1886         | Gasoline           | 1987          | 1987         | 1987         |
| Flax Seed*          | 1890          | 1890         | 1890         | Live Hog           | 1987          | 1987         | 1987         |
| Dry Salted Bellies* | 1924          | 1929         | 1924         | Oil (Brent)        | 1988          | 1988         | 1989         |
| Soybeans            | 1937          | 1937         | 1937         | Copper             | 1989          | 1989         | 1994         |
| Cotton              | 1938          | 1938         | 1938         | Natural Gas        | 1990          | 1990         | 1990         |
| Soy Bean Oil        | 1950          | 1950         | 1950         | Aluminum           | 1997          | 1997         | 1997         |
| Soy Bean Meal       | 1951          | 1951         | 1951         | Nickel             | 1997          | 1997         | 1997         |
| Sugar               | 1961          | 1964         | 1968         | Zinc               | 1997          | 1997         | 1997         |
| Cattle              | 1965          | 1965         | 1965         |                    |               |              |              |

| <b>FX</b> | <b>Return</b> | <b>Value</b> | <b>Carry</b> |
|-----------|---------------|--------------|--------------|
| GBP       | 1800          | 1871         | 1800         |
| FRF*      | 1800          | 1871         | 1800         |
| NLG*      | 1814          | 1871         | 1814         |
| NOK       | 1819          | 1871         | 1819         |
| CHF       | 1831          | 1871         | 1831         |
| DEM*      | 1832          | 1871         | 1832         |
| AUD       | 1834          | 1871         | 1834         |
| SEK       | 1836          | 1871         | 1836         |
| BEF*      | 1848          | 1871         | 1848         |
| ITL*      | 1862          | 1871         | 1862         |
| JPY       | 1880          | 1880         | 1880         |
| ESP*      | 1881          | 1881         | 1881         |
| CAD       | 1900          | 1900         | 1900         |
| NZD       | 1923          | 1923         | 1923         |
| EUR       | 1999          | 1999         | 1999         |

Note: The table shows the start years of the individual return series (before data screens) and the value and carry measures of the markets included in the investment universe ordered on start dates. Covered are equity indices, 10-year maturity government bond indices, commodities, and currencies. Series indicated with a \* are no longer active at the end of our sample.

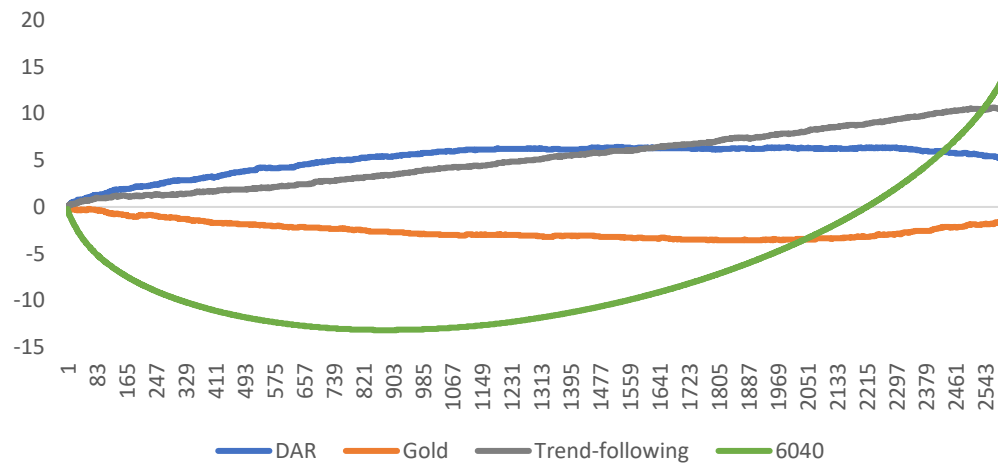
## Appendix B: Quantiles and cumulative performance sorted on 60/40 returns

**Table B1: Average performance quantiles 1800-2021**

| Quantile | Worst |       |       |       |       | Best  |       |       |       |       |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|          | 1%    | 2.5%  | 5%    | 10%   | 25%   | 25%   | 10%   | 5%    | 2.5%  | 1%    |
| 60/40    | -9.0% | -6.7% | -5.0% | -3.5% | -1.9% | 3.1%  | 4.5%  | 5.6%  | 6.9%  | 8.6%  |
| DAR4020  | 2.5%  | 1.7%  | 1.4%  | 1.0%  | 0.7%  | -0.2% | -0.5% | -0.6% | -0.8% | -1.5% |
| Gold     | -1.3% | -0.4% | -0.6% | -0.4% | -0.3% | 0.3%  | 0.4%  | 0.5%  | 0.6%  | 0.7%  |
| Trend    | 1.5%  | 1.2%  | 0.9%  | 0.5%  | 0.4%  | 0.4%  | 0.4%  | 0.2%  | 0.3%  | 0.3%  |

Note: We sort returns on the worst and best 1%, 2.5%, 5%, 10% and 25% months for the 60/40, and then compute the average monthly performance over these months for 60/40, DAR, Gold, and trend-following (Trend). Returns are scaled to 5% volatility per annum based on its ex-post realized volatility over the sample period to make them comparable. All returns are in excess of cash.

**Figure B1: Cumulative performance conditional on sorted 60/40 returns**



Note: The 60/40 monthly returns are sorted from the most negative to most positive. We then compute the cumulative performance of 60/40 (green line) along with all the protective strategies. DAR4020 is 40% long the factors that are ex-ante most negatively correlated with 60/40, and short the factors that are most positively correlated. Trend-following is long the trend factors. All protective strategies returns are scaled to 5% volatility based on its ex-post realized volatility over the sample period to make them comparable. All returns are in excess of cash. The sample period is 1805-2021, as DAR4020 needs 5 years to start, covering 2605 months.

## Appendix C: In- and out-of-sample analysis

**Table C1: In- and out-of-sample characteristics protective strategies**

|                                                          | Return<br>p.a. | Stdev<br>p.a. | IR    | >20%<br>DD | >2%<br>DD | Worst<br>10% | 60/40<br>up | 60/40<br>down |
|----------------------------------------------------------|----------------|---------------|-------|------------|-----------|--------------|-------------|---------------|
| Panel A: 1800-2021 (full sample)                         |                |               |       |            |           |              |             |               |
| DAR4020                                                  | 2.3%           | 5.0%          | 0.46  | 16.6%      | 2.8%      | 0.97%        | −0.03%      | 0.61%         |
| Gold                                                     | −0.7%          | 5.0%          | −0.14 | −2.4%      | −0.6%     | −0.40%       | 0.06%       | −0.30%        |
| Trend-following                                          | 4.8%           | 5.0%          | 0.97  | 19.3%      | 2.1%      | 0.48%        | 0.41%       | 0.38%         |
| Panel B: 1800-1984 (pre-Harvey et al. (2019) sample)     |                |               |       |            |           |              |             |               |
| DAR4020                                                  | 2.5%           | 5.1%          | 0.49  | 14.7%      | 2.9%      | 0.91%        | 0.01%       | 0.58%         |
| Gold                                                     | −1.1%          | 4.5%          | −0.24 | −5.1%      | −1.0%     | −0.55%       | 0.03%       | −0.31%        |
| Trend-following                                          | 5.1%           | 5.0%          | 1.01  | 21.7%      | 3.0%      | 0.56%        | 0.41%       | 0.44%         |
| Panel C: 1985-2021 (start Harvey et al. (2019) sample)   |                |               |       |            |           |              |             |               |
| DAR4020                                                  | 1.2%           | 4.3%          | 0.28  | 22.2%      | 2.5%      | 1.33%        | −0.23%      | 0.76%         |
| Gold                                                     | 1.0%           | 7.0%          | 0.15  | 5.5%       | 0.2%      | 0.19%        | 0.25%       | −0.23%        |
| Trend-following                                          | 3.8%           | 4.9%          | 0.76  | 12.3%      | 0.2%      | 0.07%        | 0.42%       | 0.10%         |
| Panel D: 1800-1925 (pre-Cavaglia et al. (2022) sample)   |                |               |       |            |           |              |             |               |
| DAR4020                                                  | 2.2%           | 5.0%          | 0.43  | 7.4%       | 2.1%      | 0.50%        | 0.03%       | 0.48%         |
| Gold                                                     | −2.0%          | 2.9%          | −0.69 | −12.1%     | −1.2%     | −0.59%       | −0.07%      | −0.35%        |
| Trend-following                                          | 4.3%           | 5.1%          | 0.83  | 16.8%      | 2.6%      | 0.50%        | 0.34%       | 0.38%         |
| Panel E: 1926-2021 (start Cavaglia et al. (2022) sample) |                |               |       |            |           |              |             |               |
| DAR4020                                                  | 2.4%           | 4.9%          | 0.48  | 22.1%      | 3.4%      | 1.41%        | −0.10%      | 0.78%         |
| Gold                                                     | 1.0%           | 6.8%          | 0.14  | 3.4%       | −0.2%     | −0.35%       | 0.24%       | −0.23%        |
| Trend-following                                          | 5.6%           | 4.8%          | 1.16  | 20.8%      | 1.7%      | 0.56%        | 0.50%       | 0.39%         |

Note: We examine protecting the 60/40 equities/bonds reference portfolio. Protective strategy returns are scaled to 5% volatility based on its ex-post realized volatility over the sample period in panel A, and this is not adjusted for other panels which look at subperiods. Only the returns in the column ‘return p.a.’ are annualized. The subsamples split the 1800-2021 sample based on the start date of Harvey et al. which is 1985, or the start date of Cavaglia et al. which is 1926. The Information Ratio (IR) is the ratio for the return and volatility (‘Stdev p.a.’). We look at average cumulative performance over cash over all drawdowns in excess of 20% (‘>20% DD’) or 2% (‘>2% DD’), the percentile of worst 10% returns (‘Worst 10%’), and the average monthly performance in months that 60/40 has a positive return (‘60/40 up’) or a negative return (‘60/40 down’).

## **Appendix D: The lookback horizon of trend-following**

In the main body of the text, we have defined trend-following based on past 12 months minus one month return signals, following BSV. This choice is motivated by a combination of previous results, practice in the literature, and the depth of our sample. For example, Moskowitz, Ooi and Pedersen (2012) show that past 12 months trends generally offer the highest risk-adjusted alpha across asset classes, while Hurst, Ooi and Pedersen (2017) find that 12 months trend following signals generally outperform shorter lookbacks when skipping the most recent month, a choice we make in this study to alleviate the potential impact of spurious short-term autocorrelations in asset returns. At the same time, different lookback windows can offer varying effectiveness in providing downside protection properties. For example, Harvey et al. (2019) find that following past three months equity trends outperforms past one month or past 12 months trends in S&P500 drawdowns between 1985 and 2018, while 12 months trend tends to outperform when considering recessionary states. Goulding et al. (2023b) show that across asset classes 12 months trend-following tends to outperform past 1 month trends on average and during turning points.

Another common practice in the industry and literature (e.g., Babu et al. 2020) is to blend shorter- and longer-term trend windows. A drawdown generally is a turning point, and trend-following strategies may have the wrong positions initially, thereby exacerbating early-stage drawdowns. At the same time, short term market corrections may cause faster trend variants to go short, even though the underlying market trend is still positive. Combining different trend speeds might benefit overall performance or defensiveness, as fast and slow variants dampen each other's positioning during turning points.

To examine the robustness of defensive properties of trend-following strategies over our deep sample, we consider different variants of trend-following strategies in this Appendix. We consider the 12 months speed considered above, as well as faster speeds of six months and three months lookback. We further consider a simple combination of slow and fast trend signals based on the equal weighted combination of three months and 12 months trend lookback windows. As above, note that the last month is always excluded related to potential liquidity concerns in earlier parts of our sample. Table D1 contains the results for the full sample (panel A) and for the Cavaglia et al. (2020) sample (panel B).

The results, presented in Table D1, reveal that the slower trend variant with a 12 months lookback performs best, and is superior over shorter trend lookback windows. Further, defensive properties of the faster trend variants tend to be worse, whereas one might have expected a quicker trend measure to do better in down states. Performance is especially worse in the most severe drawdowns. These results are similar over the more recent Cavaglia et al. sample period. On average, using a 12-months trend window tends to yield better defensive properties than shorter trend lookback windows over our deep sample since 1800. The last row of panel A and panel B contains results for the 50-50 combination of 12 months and three months trend. In general, results are similar to the 12 months trend in terms of unconditional performance and defensive properties.

Overall, the defensive properties of trend-following strategies are present across lookback windows, and generally best for longer-term or blended trend windows.<sup>14</sup> These results compare with substantial better downside properties for DAR4020 or the combination of DAR4020 and trend-following.

**Table D1: Trend variants**

|                    | Lookback | Return<br>p.a. | Stdev<br>p.a. | IR   | Corr<br>60/40 | >20%<br>DD | Worst<br>10% | 60/40<br>up | 60/40<br>down |
|--------------------|----------|----------------|---------------|------|---------------|------------|--------------|-------------|---------------|
| Panel A: 1800-2021 |          |                |               |      |               |            |              |             |               |
|                    | 12M-1M   | 4.8%           | 5.0%          | 0.97 | -0.05         | 19.3%      | 0.48%        | 0.41%       | 0.38%         |
|                    | 6M-1M    | 3.9%           | 5.0%          | 0.78 | -0.03         | 17.2%      | 0.45%        | 0.32%       | 0.33%         |
|                    | 3M-1M    | 2.4%           | 5.0%          | 0.49 | 0.01          | 4.0%       | 0.20%        | 0.20%       | 0.22%         |
|                    | 12M + 3M | 4.9%           | 5.0%          | 0.97 | -0.05         | 18.7%      | 0.52%        | 0.40%       | 0.41%         |
| Panel B: 1926-2020 |          |                |               |      |               |            |              |             |               |
|                    | 12M-1M   | 5.8%           | 5.0%          | 1.16 | -0.01         | 18.9%      | 0.48%        | 0.54%       | 0.39%         |
|                    | 6M-1M    | 5.1%           | 5.0%          | 1.02 | -0.01         | 16.8%      | 0.42%        | 0.46%       | 0.37%         |
|                    | 3M-1M    | 4.0%           | 5.0%          | 0.79 | 0.04          | 8.8%       | 0.11%        | 0.38%       | 0.24%         |
|                    | 12M + 3M | 6.0%           | 5.0%          | 1.20 | -0.03         | 19.4%      | 0.53%        | 0.53%       | 0.44%         |

Note: The table shows the results of different lookback windows of trend-following across asset classes. ‘12M + 3M’ combines a fast (3M-1M) and slow (12M-1M) trend lookback windows in equal measure. The following columns show the stand-alone performance: annualized return and volatility (Stdev) and their ratio (IR). The other columns show the ex-post realized correlation with 60/40, average cumulative performance in 60/40 drawdowns deeper than 20% and average monthly performance in the worst decile for 60/40 as well as conditional on 60/40 being positive (‘up’) or negative (‘down’). Panel A includes the full sample 1800-2021. Panel B aligns to the Cavaglia et al. (2022) sample at 1926-2020/4. Each protective strategy return is scaled to a 5% volatility based on its ex-post realized volatility over the sample period and in excess of cash.

<sup>14</sup> To improve trend-following strategies around turning points, Goulding, Harvey, and Mazzoleni, (2023a, 2023b) also propose a dynamic approach that alternates the weights to fast and slow trend signals. More specifically, they consider four states of the world; bear (fast and slow negative), bull (fast and slow positive), rebound (fast positive, slow negative) and correction (fast negative, slow positive), where for each state fast and slow trend strategies may be deployed. Interestingly, Goulding, Harvey, and Mazzoleni (2023a) show that dynamically combining these fast and slow trend measures tends to improve trend-following performance in equity indices, most importantly by (i) keeping the positions of trend-following strategies fully long in the ‘correction’ state, when due to the short term selloff quicker trend rules would go neutral or short, and (ii) in the ‘rebound’ state, their optimal weighting increases the tilt to fast strategies over slow ones, also resulting in more net long positions. Positions in bull and bear environments are unchanged. Given the main benefit of the approach lies in overruling short or neutral positions with more long positions, this approach has more difficulties in improving defensiveness. It adds through harvesting more equity market risk premia, thereby increasing the overall return profile. Goulding et al. (2023b) show that this approach also benefits a multi-asset trend-following strategy. That said, their findings reveal the benefit is mainly concentrated to the 2009-2019 period.